

Yield and Plastic Flow

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Introduction

In our overview of the tensile stress-strain curve in Module 4, we described yield as a permanent molecular rearrangement that begins at a sufficiently high stress, denoted σ_Y in Fig. 1. The yielding process is very material-dependent, being related directly to molecular mobility. It is often possible to control the yielding process by optimizing the materials processing in a way that influences mobility. General purpose polystyrene, for instance, is a weak and brittle plastic often credited with giving plastics a reputation for shoddiness that plagued the industry for years. This occurs because polystyrene at room temperature has so little molecular mobility that it experiences brittle fracture at stresses less than those needed to induce yield with its associated ductile flow. But when that same material is blended with rubber particles of suitable size and composition, it becomes so tough that it is used for batting helmets and ultra-durable children's toys. This magic is done by control of the yielding process. Yield control to balance strength against toughness is one of the most important aspects of materials engineering for structural applications, and all engineers should be aware of the possibilities.

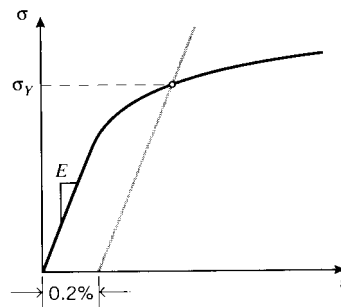


Figure 1: Yield stress σ_Y as determined by the 0.2% offset method.

Another important reason for understanding yield is more prosaic: if the material is not allowed to yield, it is not likely to fail. This is not true of brittle materials such as ceramics that fracture before they yield, but in most of the tougher structural materials no damage occurs before yield. It is common design practice to size the structure so as to keep the stresses in the elastic range, short of yield by a suitable safety factor. We therefore need to be able to predict

when yielding will occur in general multidimensional stress states, given an experimental value of σ_Y .

Fracture is driven by *normal* stresses, acting to separate one atomic plane from another. Yield, conversely, is driven by *shearing* stresses, sliding one plane along another. These two distinct mechanisms are illustrated in Fig. 2. Of course, bonds must be broken during the sliding associated with yield, but unlike in fracture are allowed to reform in new positions. This process can generate substantial change in the material, even leading eventually to fracture (as in bending a metal rod back and forth repeatedly to break it). The “plastic” deformation that underlies yielding is essentially a viscous flow process, and follows kinetic laws quite similar to liquids. Like flow in liquids, plastic flow usually takes place without change in volume, corresponding to Poisson’s ratio $\nu = 1/2$.

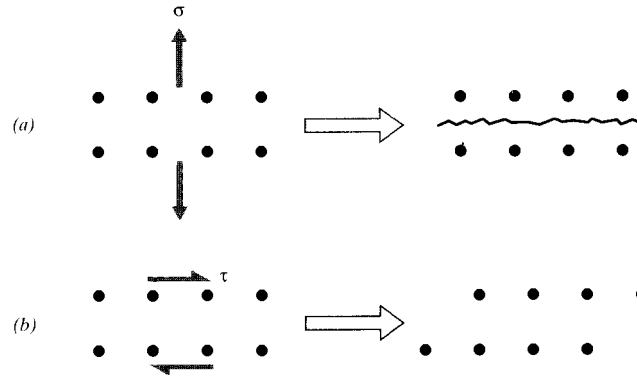


Figure 2: Cracking is caused by normal stresses (a), sliding is caused by shear stresses (b).

Multiaxial stress states

The yield stress σ_Y is usually determined in a tensile test, where a single uniaxial stress acts. However, the engineer must be able to predict when yield will occur in more complicated real-life situations involving multiaxial stresses. This is done by use of a *yield criterion*, an observation derived from experimental evidence as to just what it is about the stress state that causes yield. One of the simplest of these criteria, known as the maximum shear stress or *Tresca* criterion, states that yield occurs when the maximum shear stress reaches a critical value $\tau_{max} = k$. The numerical value of k for a given material could be determined directly in a pure-shear test, such as torsion of a circular shaft, but it can also be found indirectly from the tension test as well. As shown in Fig. 3, Mohr’s circle shows that the maximum shear stress acts on a plane 45° away from the tensile axis, and is half the tensile stress in magnitude; then $k = \sigma_Y/2$.

In cases of plane stress, Mohr’s circle gives the maximum shear stress *in that plane* as half the difference of the principal stresses:

$$\tau_{max} = \frac{\sigma_{p1} - \sigma_{p2}}{2} \quad (1)$$

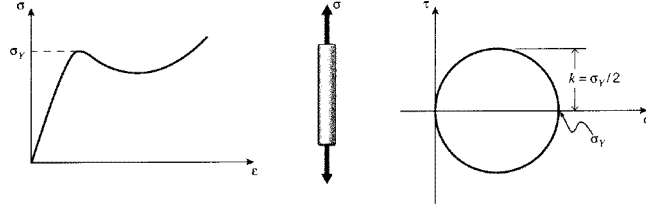


Figure 3: Mohr's circle construction for yield in uniaxial tension.

Example 1

Using $\sigma_{p1} = \sigma_\theta = pr/b$ and $\sigma_{p2} = \sigma_z = pr/2b$ in Eqn. 1, the shear stress in a cylindrical pressure vessel with closed ends is¹

$$\tau_{max,\theta z} = \frac{1}{2} \left(\frac{pr}{b} - \frac{pr}{2b} \right) = \frac{pr}{4b}$$

where the θz subscript indicates a shear stress in a plane tangential to the vessel wall. Based on this, we might expect the pressure vessel to yield when

$$\tau_{max,\theta z} = k = \frac{\sigma_Y}{2}$$

which would occur at a pressure of

$$p_Y = \frac{4b\tau_{max,\theta z}}{r} \stackrel{?}{=} \frac{2b\sigma_Y}{r}$$

However, this analysis is in error, as can be seen by drawing Mohr's circles not only for the θz plane but for the θr and rz planes as well as shown in Fig. 4.

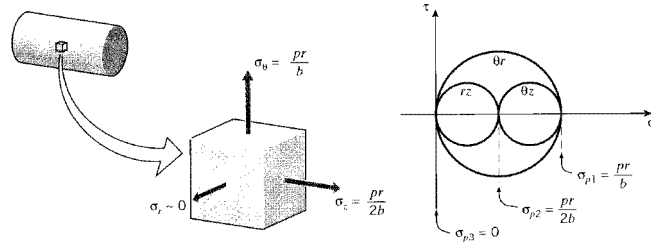


Figure 4: Principal stresses and Mohr's circle for closed-end pressure vessel

The shear stresses in the θr plane are seen to be twice those in the θz plane, since in the θr plane the second principal stress is zero:

$$\tau_{max,\theta r} = \frac{1}{2} \left(\frac{pr}{b} - 0 \right) = \frac{pr}{2b}$$

Yield will therefore occur in the θr plane at a pressure of $b\sigma_Y/r$, half the value needed to cause yield in the θz plane. Failing to consider the shear stresses acting in this third direction would lead to a seriously underdesigned vessel.

Situations similar to this example occur in plane stress whenever the principal stresses in the xy plane are of the same sign (both tensile or both compressive). The maximum shear stress,

¹See Module 6.

which controls yield, is half the *difference* between the principal stresses; if they are both of the same sign, an even larger shear stress will occur on the perpendicular plane containing the larger of the principal stresses in the xy plane.

This concept can be used to draw a “yield locus” as shown in Fig. 5, an envelope in σ_1 - σ_2 coordinates outside of which yield is predicted. This locus obviously crosses the coordinate axes at values corresponding to the tensile yield stress σ_Y . In the I and III quadrants the principal stresses are of the same sign, so according to the maximum shear stress criterion yield is determined by the difference between the larger principal stress and zero. In the II and IV quadrants the locus is given by $\tau_{max} = |\sigma_1 - \sigma_2|/2 = \sigma_Y/2$, so $\sigma_1 - \sigma_2 = \text{const}$; this gives straight diagonal lines running from σ_Y on one axis to σ_Y on the other.

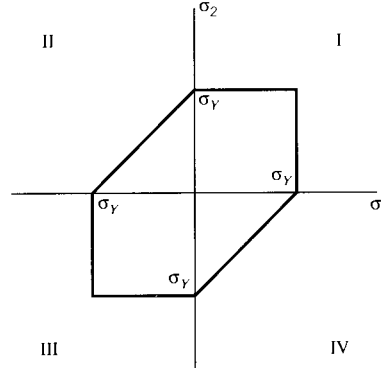


Figure 5: Yield locus for the maximum-shear stress criterion.

Example 2

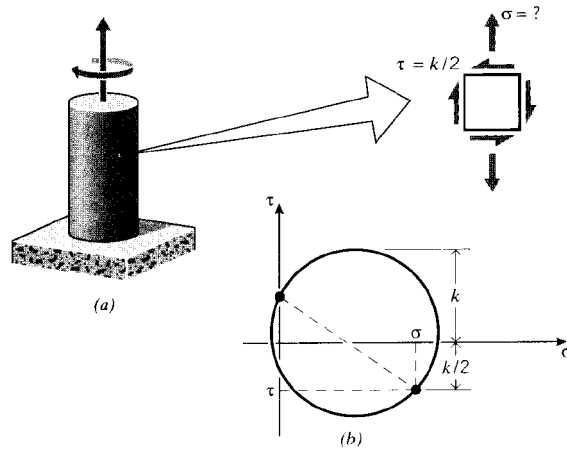


Figure 6: (a) Circular shaft subjected to simultaneous twisting and tension. (b) Mohr's circle construction.

A circular shaft is subjected to a torque of half that needed to cause yielding as shown in Fig. 6; we now ask what tensile stress could be applied simultaneously without causing yield.

A Mohr's circle is drawn with shear stress $\tau = k/2$ and unknown tensile stress σ . Using the Tresca maximum-shear yield criterion, yield will occur when σ is such that

$$\tau_{max} = k = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \left(\frac{k}{2}\right)^2}$$

$$\sigma = \sqrt{3}k$$

The Tresca criterion is convenient to use in practice, but a somewhat better fit to experimental data can often be obtained from the “von Mises” criterion, in which the driving force for yield is the strain energy associated with the deviatoric components of stress. The *von Mises stress* (also called the *equivalent* or *effective* stress) is defined as

$$\sigma_M = \sqrt{\frac{1}{2}[(\sigma_x - \sigma_y)^2 + (\sigma_x - \sigma_z)^2 + (\sigma_y - \sigma_z)^2 + 6(\tau_{xy} + \tau_{yz} + \tau_{xz})]}$$

In terms of the principal stresses this is

$$\sigma_M = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_1 - \sigma_3)^2 + (\sigma_2 - \sigma_3)^2]}$$

where the stress differences in parentheses are proportional to the maximum shear stresses on the three principal planes². (Since the quantities are squared, the order of stresses inside the parentheses is unimportant.) The Mises stress can also be written in compact form in terms of the second invariant of the deviatoric stress tensor Σ_{ij} :

$$\sigma_M = \sqrt{3\Sigma_{ij}\Sigma_{ij}/2} \quad (2)$$

It can be shown that this is proportional to the total distortional strain energy in the material, and also to the shear stress τ_{oct} on the “octahedral” plane oriented equally to the 1-2-3 axes. The von Mises stress is the driving force for damage in many ductile engineering materials, and is routinely computed by most commercial finite element stress analysis codes.

The value of von Mises stress $\sigma_{M,Y}$ needed to cause yield can be determined from the tensile yield stress σ_Y , since in tension at the yield point we have $\sigma_1 = \sigma_Y$, $\sigma_2 = \sigma_3 = 0$. Then

$$\sigma_{M,Y} = \sqrt{\frac{1}{2}[(\sigma_Y - 0)^2 + (\sigma_Y - 0)^2 + (0 - 0)^2]} = \sigma_Y$$

Hence the value of von Mises stress needed to cause yield is the same as the simple tensile yield stress.

The shear yield stress k can similarly be found by inserting the principal stresses corresponding to a state of pure shear in the Mises equation. Using $k = \sigma_1 = -\sigma_3$ and $\sigma_2 = 0$, we have

$$\sqrt{\frac{1}{2}[(k - 0)^2 + (k + k)^2 + (0 - k)^2]} = \sqrt{\frac{6k^2}{2}} = \sigma_Y$$

$$k = \frac{\sigma_Y}{\sqrt{3}}$$

²Some authors use a factor other than 1/2 within the radical. This is immaterial, since it will be absorbed by the calculation of the critical value of σ_M .

Note that this result is different than the Tresca case, in which we had $k = \sigma_Y/2$.

The von Mises criterion can be plotted as a yield locus as well. Just as the Tresca criterion, it must pass through σ_Y on each axis. However, it plots as an ellipse rather than the prismatic shape of the Tresca criterion (see Fig. 7).

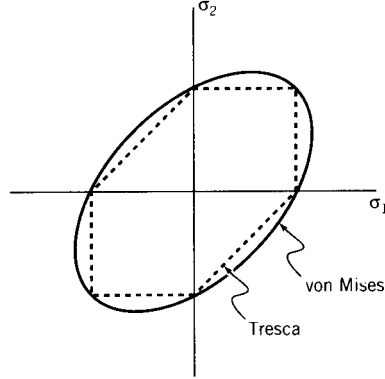


Figure 7: Yield locus for the von Mises criterion.

Effect of hydrostatic pressure

Since in the discussion up to now yield has been governed only by shear stress, it has not mattered whether a uniaxial stress is compressive or tensile; yield occurs when $\sigma = \pm\sigma_Y$. This corresponds to the hydrostatic component of the stress $-p = (\sigma_x + \sigma_y + \sigma_z)/3$ having no influence on yield, which is observed experimentally to be valid for slip in metallic systems. Polymers, however, are much more resistant to yielding in compressive stress states than in tension. The atomistic motions underlying slip in polymers can be viewed as requiring “free volume” as the molecular segments move, and this free volume is diminished by compressive stresses. It is thus difficult to form solid polymers by deformation processing such as stamping and forging in the same way steel can be shaped; this is one reason the vast majority of automobile body panels continue to be made of steel rather than plastic.

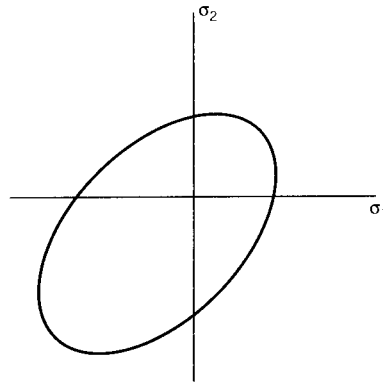


Figure 8: Effect of pressure on the von Mises yield envelope.

This dependency on hydrostatic stress can be modeled by modifying the yield criterion to state that yield occurs when

$$\tau_{max} \text{ or } \sigma_M \geq \tau_0 + Ap \quad (3)$$

where τ_0 and A are constants. As p increases (the hydrostatic component of stress becomes more positive) the shear stress needed for yield becomes greater as well, since there is less free volume and more hindrance to molecular motion. The effect of this modification is to slide the von Mises ellipse to extend less into the I quadrant and more into the III quadrant as shown in Fig. 8. This shows graphically that greater stresses are needed for yield in compression, and lesser stresses in tension.

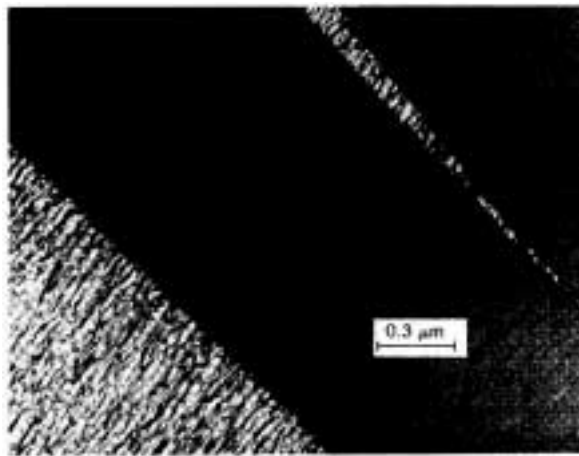


Figure 9: A craze in polystyrene (from R. Kambour, “Crazing,” *Encyclopedia of Polymer Science and Engineering*, Wiley-Interscience, 1991).

Several amorphous glassy polymers — notably polystyrene, polymethylmethacrylate, and polycarbonate — are subject to a yield mechanism termed “crazing” in which long elongated voids are created within the material by a tensile cavitation process. Figure 9 shows a craze in polystyrene, grown in plasticizing fluid near T_g . The voids, or crazes, are approximately 1000Å thick and microns or more in length, and appear visually to be much like conventional cracks. They differ from cracks, however, in that the broad faces of the crazes are spanned by a great many elongated fibrils that have been drawn from the polymer as the craze opens. The fibril formation requires shear flow, but the process is also very dependent on free volume. A successful multiaxial stress criterion for crazing that incorporates both these features has been proposed³ of the form

$$\sigma_1 - \sigma_2 = A(T) + \frac{B(T)}{\sigma_1 + \sigma_2}$$

The left hand side of this relation is proportional to the shear stress, and the denominator in the second term on the right hand side is related to the hydrostatic component of the stress. As the hydrostatic tension increases, the shear needed to cause crazing decreases. The parameters A and B are adjustable, and both depend on temperature. This relation plots as a batwing on the yield locus diagram as seen in Fig. 10, approaching a 45° diagonal drawn through the II

³S. Sternstein and L. Ongchin, *Polymer Preprints*, **10**, 1117, 1969.

and IV quadrants. Crazing occurs to the right of the curve; note that crazing never occurs in compressive stress fields.

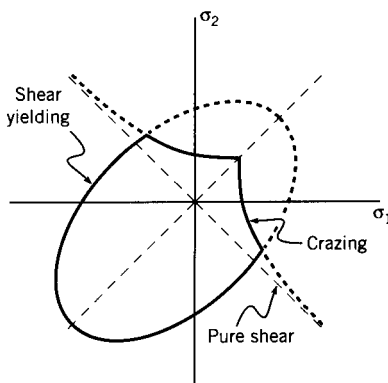


Figure 10: The Sternstein envelopes for crazing and pressure-inhibited shear yielding.

Crazing is a yield mechanism, but it also precipitates brittle fracture as the craze height increases and the fibrils are brought to rupture. The point where the craze locus crosses the shear yielding locus is therefore a type of mechanically induced ductile-brittle transition, as the failure mode switches from shear yielding to craze embrittlement. Environmental agents such as acetone that expand the free volume in these polymers greatly exacerbate the tendency for craze brittleness. Conversely, modifications such as rubber particle inclusions that stabilize the crazes and prevent them from becoming true cracks can provide remarkable toughness. Rubber particles not only stabilize crazes, they also cause a great increase in the number of crazes, so the energy absorption of craze formation can add to the toughness as well. This is the basis of the “high impact polystyrene,” or HIPS, mentioned at the outset of this chapter.

Effect of rate and temperature

The yield process can be viewed as competing with fracture, and whichever process has the lowest stress requirements will dominate. As the material is made less and less mobile, for instance by lowering the temperature or increasing the number and tightness of chemical bonds, yielding becomes more and more difficult. The fracture process is usually much less dependent on mobility. Both yield and fracture stresses usually increase with decreasing temperature, but yield is more temperature-dependent (see Fig. 11). This implies that below a critical temperature (called the ductile-brittle transition temperature T_{DB}) the material will fracture before it yields. Several notable failures in ships and pipelines have occurred during winter temperatures when the steels used in their manufacture were stressed below their T_{DB} and were thus unable to resist catastrophic crack growth. In polymers, the ductile-brittle transition temperature is often coincident with the glass transition temperature. Clearly, we need an engineering model capable of showing how yield depends on temperature, and one popular approach is outlined below.

Yield processes are thermally activated, stress driven motions, much like the flow of viscous liquids. Even without going into much detail as to the specifics of the motions, it is possible to write down quite effective expressions for the dependency of these motions on strain rate and temperature. In the Eyring view of thermally activated processes, an energy barrier E_Y^* must be overcome for the motion to proceed. (We shall use the asterisk superscript to indicate activation

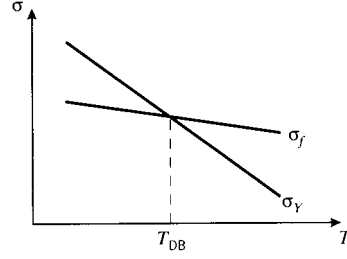


Figure 11: Schematic illustration of the temperature dependence of yield and fracture stress.

parameters, and the Y subscript here indicates the yield process.) A stress acts to lower the barrier when it acts in the direction of flow, and to raise it when it opposes the flow.

Consider now a constant strain rate test ($\dot{\epsilon} = \text{const}$), in which the stress rises until yield occurs at $\sigma = \sigma_Y$. At the yield point we have $d\sigma/d\epsilon = 0$, so a fluidlike state is achieved in which an increment of strain can occur without a corresponding incremental increase in stress. Analogously with rate theories for viscous flow, an Eyring rate equation can be written for the yielding process as

$$\dot{\epsilon} = \dot{\epsilon}_0 \exp \frac{-(E_Y^* - \sigma_Y V^*)}{kT} \quad (4)$$

Here k is Boltzman's constant and V^* is a factor governing the effectiveness of the stress in reducing the activation barrier. It must have units of volume for the product $\sigma_Y V^*$ to have units of energy, and is called the "activation volume" of the process. Taking logs and rearranging,

$$\frac{\sigma_Y}{T} = \frac{E_Y^*}{V^* T} + \left(\frac{k}{V^*} \right) \ln \left(\frac{\dot{\epsilon}}{\dot{\epsilon}_0} \right)$$

Hence plots of σ_Y/T versus $\ln \dot{\epsilon}$ should be linear with a slope k/V^* as seen in Fig. 12, from which the activation volume may be computed. The horizontal spacing between two lines at differing temperatures T_1 and T_2 gives the activation energy:

$$E_Y^* = \frac{k (\ln \dot{\epsilon}^{T_2} - \ln \dot{\epsilon}^{T_1})}{\left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$

Apparent activation volumes in polymers are on the order of $5000\text{\AA}^3$, much larger than a single repeat unit. This is taken to indicate that yield in polymers involves the cooperative motion of several hundred repeat units.

Example 3

The yield stress for polycarbonate is reported at 60 MPa at room-temperature ($23^\circ\text{C} = 296^\circ\text{K}$), and we wish to know its value at 0°C (273°K), keeping the strain rate the same.

This can be accomplished by writing Eqn. 4 out twice, once for each temperature, and then dividing one by the other. The parameters $\dot{\epsilon}$ and $\dot{\epsilon}_0$ cancel, leaving

$$1 = \exp \left(\frac{E_Y^* - \sigma_Y^{273} V^*}{R(273)} - \frac{E_Y^* - \sigma_Y^{296} V^*}{R(296)} \right)$$

From the data in Fig. 12, the yield activation parameters are $E_Y^* = 309 \text{ kJ/mol}$, $V^* = 3.9 \times 10^{-3} \text{ m}^3/\text{mol}$. Using these along with $R = 8.314 \text{ J/mol}$ and $\sigma_Y^{296} = 60 \times 10^6 \text{ N/m}^2$, we have

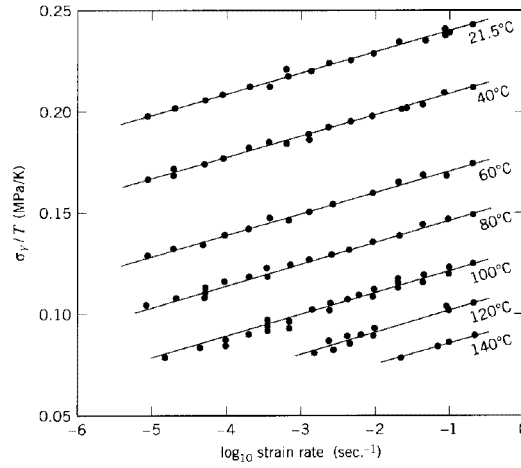


Figure 12: Eyring plot showing dependence of yield strength on temperature and strain rate in polycarbonate (from N.G. McCrum, C.P. Buckley and C.B. Bucknall, *Principles of Polymer Engineering*, Oxford University Press, 1988).

$$\sigma_Y^{273} = 61.5 \text{ MPa}$$

Continuum plasticity

Plasticity theory, which seeks to determine stresses and displacements in structures all or part of which have been stressed beyond the yield point, is an important aspect of solid mechanics. The situation is both materially and geometrically nonlinear, so it is not a trivial undertaking. However, in such areas as metal forming, plasticity theory has provided valuable insight. We will outline only a few aspects of this field in the following paragraphs, to introduce some of the fundamental concepts that the reader can extend in future study.

Plastic deformation

A useful idealization in modeling plastic behavior takes the material to be linearly elastic up to the yield point as shown in Fig. 13, and then “perfectly plastic” at strains beyond yield. Strains up to yield (the line between points *a* and *b*) are recoverable, and the material unloads along the same elastic line it followed during loading; this is conventional elastic response. But if the material is strained beyond yield (point *b*), the “plastic” straining beyond *b* takes place at constant stress and is unrecoverable. If the material is strained to point *c* and then unloaded, it follows the path *cd* (a line parallel to the original elastic line *ab*) rather than returning along *cba*. When the stress has been brought to zero (point *d*), the plastic strain *ad* remains as a *residual* strain.

Plastic deformation can generate “residual” stresses in structures, internal stresses that remain even after the external loads are removed. To illustrate this, consider two rods having different stress-strain curves, connected in parallel (so their strains are always equal) as shown in Fig. 14. When the rods are strained up to the yield point of rod *B* (point *a* on the strain

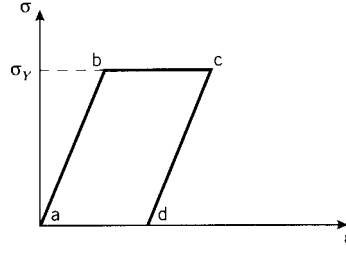


Figure 13: The elastic-perfectly plastic idealization of plastic deformation.

axis), rod *A* will have experienced an amount of permanent plastic deformation ϵ^p . When the applied load is removed, rod *B* unloads along its original stress-strain curve, but rod *A* follows a path parallel to its original elastic line. When rod *A* reaches zero stress (point *b*), rod *B* will still be in tension (point *c*). In order for the load transmitted by the rods together to come to zero, rod *B* will pull rod *A* into compression until $-\sigma_B = \sigma_A$ as indicated by points *d* and *e*. Residual stresses are left in the rods, and the assembly as a whole is left with a residual tensile strain.

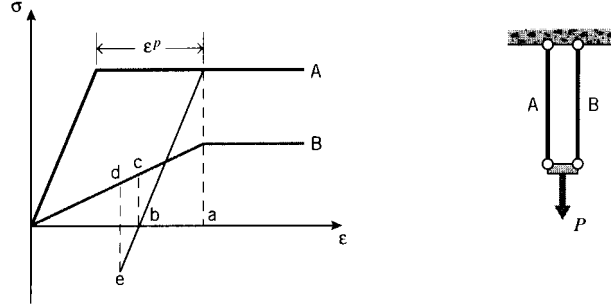


Figure 14: Plastic deformation of two-bar assembly.

Compressive residual stress can be valuable if the structure must bear tensile loads. Similarly to how rapid quenching can be used to make safety glass by putting the surfaces in compression, plastic deformation can be used to create favorable compressive stresses. One famous such technique is called “autofrettage;” this is a method used to strengthen cannon barrels against bursting by pressurizing them from the inside so as to bring the inner portion of the barrel into the plastic range. When the pressure is removed, the inner portions are left with a compressive residual stress just as with bar *A* in the above example.

Wire drawing

To quantify the plastic flow process in more detail, consider next the “drawing” of wire⁴, in which wire is pulled through a reducing die so as to reduce its cross-sectional area from A_0 to A as shown in Fig. 15. Since volume is conserved during plastic deformation, this corresponds to an axial elongation of $L/L_0 = A_0/A$. Considering the stress state to be simple uniaxial tension, we have

⁴G.W. Rowe, *Elements of Metalworking Theory*, Edward Arnold, London, 1979.

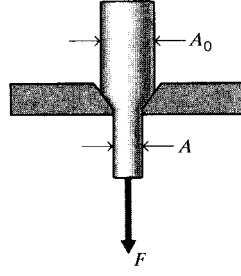


Figure 15: Wire drawing.

$$\sigma_1 = \sigma_Y, \quad \sigma_2 = \sigma_3 = 0$$

where 1 denotes the direction along the wire and 2 and 3 are the transverse directions. The work done in stretching the wire by an increment of length dL , per unit volume of material, is

$$dU = \frac{dW}{AL} = \frac{\sigma_Y A dL}{AL}$$

Integrating this from L_0 to L to obtain the total work:

$$U = \int_{L_0}^L dU = \frac{F dL}{AL} = \sigma_Y \ln \frac{L}{L_0}$$

The quantity $\ln(L/L_0)$ is the *logarithmic strain* ϵ_T introduced in Module 4 (Stress-Strain Curves).

Example 4

The logarithmic strain can be written in terms of either length increase or area reduction, due to the constancy of volume during plastic deformation: $\epsilon_T = \ln(L/L_0) = \ln(A_0/A)$. In terms of diameter reduction, the relation $A = \pi d^2/4$ leads to

$$\epsilon_T = \ln \left(\frac{\pi d_0^2/4}{\pi d^2/4} \right) = 2 \ln \left(\frac{d_0}{d} \right)$$

Taking the pearlite cell size to shrink commensurately with the diameter, we expect the wire strength σ_f to vary according to the Hall-Petch relation with $1/\sqrt{d}$. The relation between wire strength and logarithmic drawing strain is then

$$\sigma_f \propto \frac{\exp(\epsilon_T/4)}{\sqrt{d_o}}$$

The work done by the constant pulling force F in drawing an initial length L_0 of wire to a new length L is $W = FL$. This must equal the work per unit volume done in the die, multiplied by the total volume of wire:

$$FL = (AL) \sigma_Y \ln \frac{L}{L_0}$$

Written in terms of area reduction, this is

$$F = A \sigma_Y \ln \frac{A_0}{A}$$

This simple result is useful in estimating the requirements of wire drawing, even though it neglects the actual complicated flow field within the die and the influence of friction at the die walls. Both friction at the surface and constraints to flow within the field raise the force needed in drawing, but the present analysis serves to establish a lower-limit approximation. It is often written in terms of the drawing stress $\sigma_1 = F/A$ and the area reduction ratio $r = (A_0 - A)/A_0 = 1 - (A/A_0)$:

$$\sigma_1 = \sigma_Y \ln \frac{1}{1-r}$$

Note that the draw stress for a small area reduction is less than the tensile yield stress. In fact, the maximum area reduction that can be achieved in a single pass can be estimated by solving for the value of r which brings the draw stress up to the value of the yield stress, which it obviously cannot exceed. This calculation gives

$$\ln \frac{1}{1-r_{max}} = 1 \Rightarrow r_{max} = 1 - \frac{1}{e} = 0.63$$

Hence the maximum area reduction is approximately 63%, assuming perfect lubrication at the die. This lower-bound treatment gives an optimistic result, but is not far from the approximately 50% reduction often used as a practical limit. If the material hardens during drawing, the maximum reduction can be slightly greater.

Slip-line fields

In cases of plane strain, there is a graphical technique called *slip-line theory*⁵ which permits a more detailed examination of plastic flow fields and the loads needed to create them. Friction and internal flow constraints can be included, so upper-bound approximations are obtained that provide more conservative estimates of the forces needed in deformation. Considerable experience is needed to become proficient in this method, but the following will outline some of the basic ideas.

Consider plane strain in the 1-3 plane, with no strain in the 2-direction. There is a Poisson stress in the 2-direction, given by

$$\epsilon_2 = 0 = \frac{1}{E}[\sigma_2 - \nu(\sigma_1 + \sigma_3)]$$

Since $\nu = 1/2$ in plastic flow,

$$\sigma_2 = \frac{1}{2}(\sigma_1 + \sigma_3)$$

The hydrostatic component of stress is then

$$p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3) = \frac{1}{2}(\sigma_1 + \sigma_3) = \sigma_2$$

Hence the Poisson stress σ_2 in the zero-strain direction is the average of the other two stresses σ_1 and σ_3 , and also equal to the hydrostatic component of stress. The stress state can be specified in terms of the maximum shear stress, which is just k during plastic flow, and the superimposed hydrostatic pressure p :

$$\sigma_1 = -p + k, \quad \sigma_2 = -p \quad \sigma_3 = -p - k$$

⁵W. Johnson and P.B. Mellor, *Plasticity for Mechanical Engineers*, Van Nostrand Co., New York, 1962.

Since the shear stress is equal to k everywhere, the problem is one of determining the directions of k (the direction of maximum shear, along which slip occurs), and the magnitude of p .

The graphical technique involves sketching lines that lie along the directions of k . Since maximum shear stresses act on two orthogonal planes, there will be two sets of these lines, always perpendicular to one another and referred to as α -lines and β -lines. The direction of these lines is specified by an inclination angle ϕ . Any convenient inclination can be used for the $\phi = 0$ datum, and the identification of α - vs. β -lines is such as to make the shear stress positive according to the usual convention. As the pressure p varies from point to point, there is a corresponding variation of the angle ϕ , given by the *Hencky equations* as

$$p + 2k\phi = C_1 = \text{constant, along an } \alpha\text{-line}$$

$$p - 2k\phi = C_2 = \text{constant, along a } \beta\text{-line}$$

Hence the pressure can be determined from the curvature of the sliplines, once the constant is known.

The slip-line field must obey certain constraints at boundaries:

1. *Free surfaces*: Since there can be no stress normal to a free surface, we can put $\sigma_3 = 0$ there and then

$$p = k, \quad \sigma_1 = -p - k = -2k$$

Hence the pressure is known to be just the shear yield strength at a free surface. Furthermore, since the directions normal and tangential to the surface are principal directions, the directions of maximum shear must be inclined at 45° to the surface.

2. *Frictionless surface*: The shear stress must be zero tangential to a frictionless surface, which again means that the tangential and normal directions must be principal directions. Hence the slip lines must meet the surface at 45° . However, there will in general be a stress acting normal to the surface, so $\sigma_3 \neq 0$ and thus p will not be equal to k .
3. *Perfectly rough surface*: If the friction is so high as to prevent any tangential motion at the surface, the shearing must be maximum in a direction that is also tangential to the surface. One set of slip lines must then be tangential to the surface, and the other set normal to it.

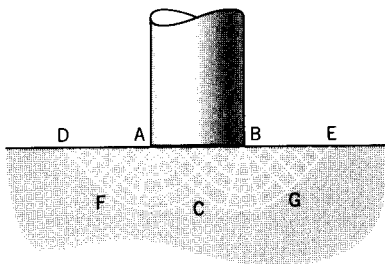


Figure 16: Slip-line construction for a flat indentation.

Consider a flat indenter of width b being pressed into a semi-infinite block, with negligible friction (see Fig. 16). Since the sliplines must meet the indenter surface at 45° , we can draw a

triangular flow field ABC . Since all lines in this region are straight, there can be no variation in the pressure p , and the field is one of “constant state.” This cannot be the full extent of the field, however, since it would be constrained both vertically and laterally by rigid metal. The field must extend to the free surfaces adjacent to the punch, so that downward motion under the punch can be compensated by upward flow adjacent to it. Two more triangular regions ADF and BEG are added that satisfy the boundary conditions at free surfaces, and these are connected to the central triangular regions by “fans” AFC and BCG . Fans are very useful in slip-line constructions; they are typically centered on singularities such as points A and B where there is no defined normal to the surface.

The pressure on the punch needed to establish this field can be determined from the sliplines, and this is one of their principal uses. Since BE is a free surface, $\sigma_3 = 0$ there and $p = k$. The pressure remains constant along line EG since ϕ is unchanging, but as ϕ decreases along the curve GC (the line curves clockwise), the pressure must increase according to the Hencky equation. At point C it has rotated through $-\pi/2$ so the pressure there is

$$p_C + 2k\phi = p_C + 2k\left(-\frac{\pi}{2}\right) = \text{constant} = p_G = k$$

$$p_C = k(1 + \pi)$$

The pressure remains unchanged along lines CA and CB , so the pressure along the punch face is also $k(1 + \pi)$. The total stress acting upward on the punch face is therefore

$$\sigma_1 = p + k = 2k\left(1 + \frac{\pi}{2}\right)$$

The ratio of punch pressure to the tensile yield strength $2k$ is

$$\frac{\sigma_1}{2k} = 1 + \frac{\pi}{2} = 2.571$$

The factor 2.571 represents the increase over the tensile yield strength caused by the geometrical constraints on the flow field under the punch.

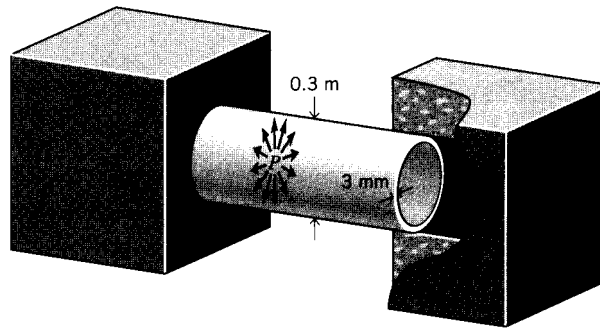
The *Brinell Hardness Test* is similar to the punch yielding scenario above, but uses a hard steel sphere instead of a flat indenter. The Brinell hardness H is calculated as the load applied to the punch divided by the projected area of the indentation. Analysis of the Brinell test differs somewhat in geometry, but produces a result not much different than that of the flat punch:

$$\frac{H}{\sigma_Y} \approx 2.8 - 2.9$$

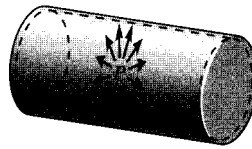
This relation is very useful in estimating the yield strength of metals by simple nondestructive indentation hardness tests.

Problems

1. An open-ended pressure vessel is constructed of aluminum, with diameter 0.3 m and wall thickness 3 mm. (Open-ended in this context means that both ends of the vessel are connected to other structural parts able to sustain pressure, as in a hose connected between two reservoirs.) Determine the internal pressure at which the vessel will yield according to the (a) Tresca and (b) von Mises criteria.

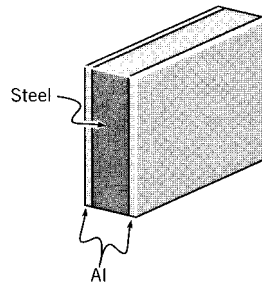


Prob. 1



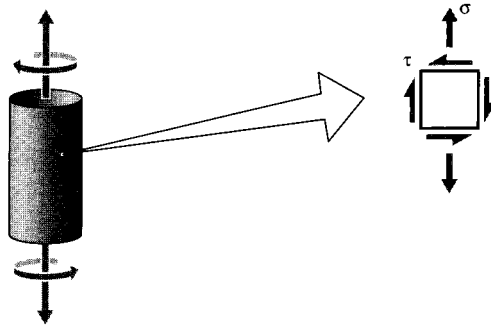
Prob. 2

2. Repeat the previous problem, but with the pressure vessel now being closed-ended.
3. A steel plate is clad with a thin layer of aluminum on both sides at room temperature, and the temperature then raised. At what temperature increase ΔT will the aluminum yield?



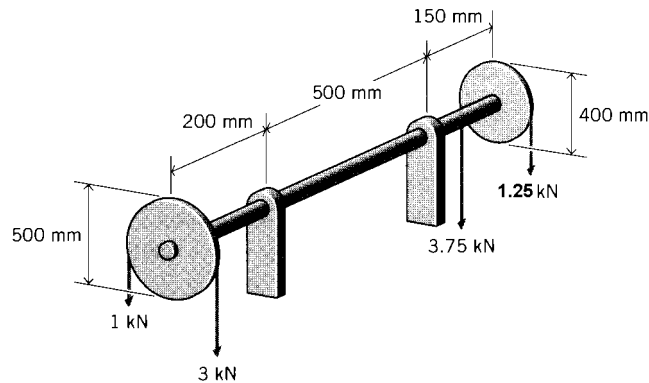
Prob. 3

4. If the temperature in the previous problem is raised 40°C beyond the value at which yielding occurs, and is thereafter lowered back to room temperature, what is the residual stress left in the aluminum?
5. Copper alloy is subjected to the stress state $\sigma_x = 100, \sigma_y = -200, \tau_{xy} = 100$ (all in MPa). Determine whether yield will occur according to the (a) Tresca and (b) von Mises criterion.
6. Repeat the previous problem, but with the stress state $\sigma_x = 190, \sigma_y = 90, \tau_{xy} = 120$ (all in MPa)
7. A thin-walled tube is placed in simultaneous tension and torsion, causing a stress state as shown here. Construct a plot of τ/σ_Y vs. σ/σ_Y at which yield will occur according to the (a) Tresca and (b) von Mises criterion. (σ_Y is the tensile yield stress.)



Prob. 7

8. A solid circular steel shaft is loaded by belt pulleys at both ends as shown. Determine the diameter of the shaft required to avoid yield according to the von Mises criterion, with a factor of safety of 2.



Prob. 8

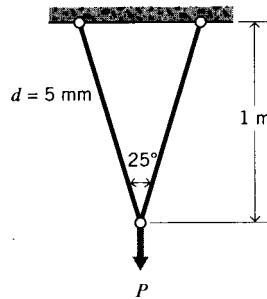
9. For polycarbonate, the kinetic parameters in Eqn. 4 are found to be $\dot{\epsilon}_0 = 448 \text{ s}^{-1}$, $E_Y^* = 309 \text{ kJ/mol}$, and $V^* = 3.9 \times 10^{-3} \text{ m}^3/\text{mol}$. Find the yield stress σ_Y at a strain rate of $\dot{\epsilon} = 10^2 \text{ s}^{-1}$ and temperature 40°C .
10. Yield stresses (in MPa) have been measured at various strain rates and temperatures as follows:

	$\dot{\epsilon} = 10^{-3} \text{ s}^{-1}$	$\dot{\epsilon} = 10^{-1} \text{ s}^{-1}$
$T = 0^\circ\text{C}$	54.1	62.7
$T = 40^\circ\text{C}$	42.3	52.1

Determine the activation volume for the yield process. What physical significance might this parameter have?

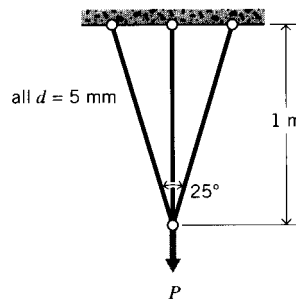
11. The yield stress of a polymer is measured to be 20 MPa at a temperature of 300K and a strain rate of 10^{-3} s^{-1} . When the strain rate is doubled from this value, the yield stress is observed to increase by 10%. What is the apparent activation volume for yield in this case?

12. Show the von Mises stress can be written in index notation as $\sigma_M = \sqrt{3\Sigma_{ij}\Sigma_{ij}/2}$
13. A sample of linear polyethylene was tested in uniaxial loading at $T = 23^\circ \text{ C}$ and $\dot{\epsilon} = 10^{-3} \text{ s}^{-1}$. The yield stress σ_Y was found to be 30.0 MPa in tension and 31.5 MPa in compression. Determine the pressure-dependency constant A in Eqn. 3.
14. A circular shaft of radius R is subjected to a torque T .
- What value of T will be just large enough to induce yielding at the outer surface?
 - As the value of T is increased beyond the level found in (a), determine the radius r_e within which the material is still in the elastic range.
 - What value of T will make the shaft fully plastic; i.e. $r_e = 0$?
15. A two-element truss frame is constructed of steel with the geometry shown. What load P can the frame support without yielding in either element?



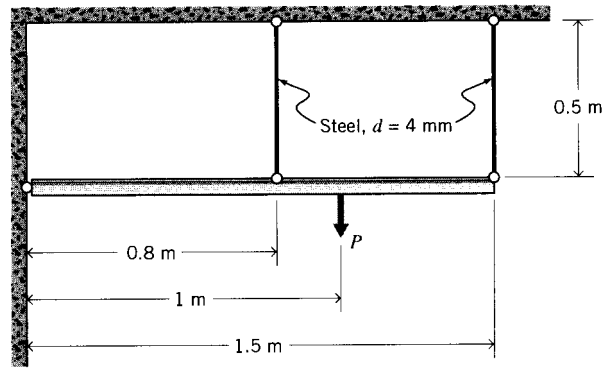
Prob. 15

16. A three-element truss frame is constructed of steel with the geometry shown. What load P can the frame support before all three elements have yielded?



Prob. 16

17. If the frame of the previous problem is loaded until all three members have yielded and the load then reduced to zero, find the residual stress in the central element.
18. A rigid beam is hinged at one end as shown and supported by two vertical rods as shown.
- What load P can the structure support before both vertical rods have yielded?
 - What is the residual stress in the vertical rods after the load has been reduced to zero?



Prob. 18

19. Estimate the drawing force required to reduce the diameter of a 0.125" aluminum rod by 50% in a wire-drawing operation.

The Dislocation Basis of Yield and Creep

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Introduction

Phenomenological treatments such those outlined in the Module on Yield and Plastic Flow (Module 20) are very useful for engineering predictions, but they provide only limited insight to the molecular mechanisms underlying yield. Molecular understanding is a higher level of insight, and also guides processing adjustments that can optimize the material. As discussed in the Module on Atomistics of Elasticity (Module 2), the high level of order present in crystalline materials lead to good atomistic models for the stiffness. Early workers naturally sought an atomistic treatment of the yield process as well. This turned out to be a much more subtle problem than might have been anticipated, and required hypothesizing a type of crystalline defect — the “dislocation” — to explain the experimentally observed results. Dislocation theory permits a valuable intuitive understanding of yielding in crystalline materials, and explains how yielding can be controlled by alloying and heat treatment. It is one of the principal triumphs of the last century of materials science.

Theoretical yield strength

In yield, atoms slide tangentially from one equilibrium position to another. The forces required to bring this about are given by the bond energy function, which is the anharmonic curve resulting from the balance of attractive and repulsive atomic forces described in Module 2. The force needed to displace the atom from equilibrium is the derivative of the energy function, being zero at the equilibrium position (see Fig. 1). As a simplifying assumption, let us approximate the force function with a harmonic expression, and write

$$\tau = \tau_{max} \sin \left(2\pi \frac{x}{a} \right)$$

where a is the interatomic spacing. The stress reaches a maximum a quarter of the distance between the two positions, dropping to zero at the metastable position midway between them. After that, the stress changes sign, meaning that force is required to hold the atom back as it tries to fall toward the new equilibrium position. Using $\gamma = x/a$ as the shear strain, the maximum shear stress τ_{max} can be related to the shear modulus G as

$$\frac{d\tau}{d\gamma} = \frac{d\tau}{dx} \frac{dx}{d\gamma} = a \frac{d\tau}{dx} = a \tau_{max} \frac{2\pi}{a} \cos \frac{2\pi x}{a}$$

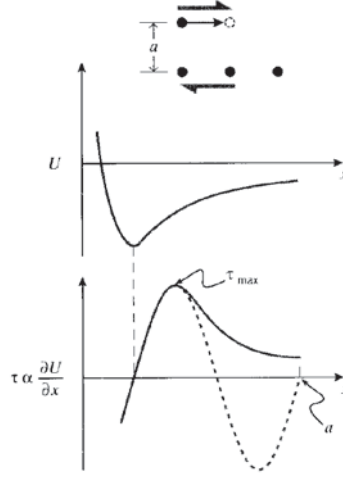


Figure 1: Atomistic energy and stress functions.

$$G = \left. \frac{d\tau}{d\gamma} \right|_{\gamma \rightarrow 0} = \tau_{max} \cdot 2\pi$$

This implies a shear stress at yield of $\tau_{max} = G/2\pi \approx G/10$, which would be on the order of 10 GPa. Measured values are 10–100 MPa, so the theoretical value is 2 to 3 orders of magnitude too large. More elaborate derivations give a somewhat smaller value for the theoretical yield stress, but still much larger than what is observed experimentally.

Edge, screw, and mixed dislocations

A rationale for the apparently low experimental values for the yield strengths of crystalline materials was proposed independently by Taylor, Polanyi and Orowan in 1934. These workers realized that it was not necessary to slip entire planes of atoms past one another to deform the material plastically, a process that would require breaking all the bonds connecting the planes simultaneously. The stress needed to do this would be very high, on the order of $G/10$ as described above. But it isn't necessary to move all the atoms at once; only a few at a time need to move, requiring a much smaller stress. Analogously to the way an inchworm moves, only those atoms lying in a plane above a single line might be displaced one atomic spacing. This would force the plane of atoms previously there into a midway position as shown in Fig. 2, creating an “extra” plane of atoms halfway between the normal equilibrium positions. The termination of this plane then constitutes a line defect in the crystal known as a *dislocation*¹.

Viewed end-on as seen in Fig. 3, it can be appreciated that the extra plane of atoms creates a region of compression near the plane but above the dislocation line, and a tensile region below it. In a “soft” crystal whose interatomic bonds are relatively compliant, the distortion extends an appreciable distance from the dislocation. Conversely, in “hard” crystals with stiffer bonds the distortion is confined to a smaller region near the dislocation. The face-centered cubic (fcc) metals such as copper and gold have close-packed planes (those with (111) Miller indices), which

¹For an extended discussion of the geometrical aspects of crystal dislocations, see S.M. Allen and E.L. Thomas, *The Structure of Materials*, John Wiley & Sons, New York, 1999.

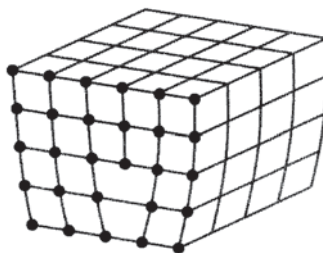


Figure 2: The edge dislocation.

corresponds to large distances between those planes. This gives rise to relatively soft interplanar bonds, so that the dislocation width is large. The dislocation width is substantially smaller in the body centered cubic (bcc) metals such as iron and steel, smaller still in ionically bonded ceramics, and even smaller in covalently bonded ceramics.

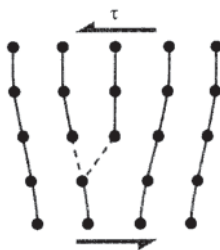


Figure 3: Dislocation motion.

The dislocation associated with this extra plane of atoms can be moved easily, since only a small adjustment in position is required to break the bonds on the next plane over and allow them to form on the originally “extra” plane. Now the third plane is the extra one, and the dislocation will have moved by one atomic position. Slip is obviously made much easier if dislocation motion is available to the material. In fact, it first appears the dislocation concept does *too* good a job in explaining crystal plasticity, since the dislocation is in a balanced metastable position and should be capable of being moved either left or right with a vanishingly small force. If this were true, the crystal would have essentially *no* shear strength.

However, as the dislocation moves it drags with it the regions of compressive and tensile distortion in the lattice around it. This is accompanied by a sort of frictional drag, giving rise to a resistance to dislocation motion known as the *Peierls* force. This force is dependent on such factors as the crystal type and the temperature, and this plays an important role in determining the material’s yield stress. As seen in Table 1, materials with wide dislocations have low Peierls forces, since the distortion is spread out over a large volume and is much less intense at its core.

Table 1 also indicates that the effect of temperature on the Peierls force is low for fcc materials having wide dislocations, and this results in a small temperature dependency of the yield stress. Conversely, materials with narrow and intense dislocation fields have high Peierls forces with a large temperature sensitivity of the yield stress, with higher temperatures facilitating dislocation mobility and thus reducing the yield strength. Among the important consequences of these factors is the dangerous tendency of steel to become brittle at low temperatures; as the temperature is lowered, the yield stress can rise to such high levels that brittle fracture

Table 1: Relationship between dislocation width and yield strength temperature sensitivity (from R.W. Hertzberg, *Deformation and Fracture Mechanics of Engineering Materials*, John Wiley & Sons, 1976).

Material	Crystal Type	Dislocation Width	Peierls Stress	Yield Strength Temperature Sensitivity
Metal	fcc	wide	very small	negligible
Metal	bcc	narrow	moderate	strong
Ceramic	ionic	narrow	large	strong
Ceramic	covalent	very narrow	very large	strong

intervenes.

Dislocations can have geometries other than the simple edge dislocation shown in Fig. 3. A more general view is provided by considering displacing a portion of the atoms in a “slip plane” $acfg$ a distance $\bar{b}$, as shown in Fig. 4. The vector $\bar{b}$ is also a measure of the magnitude and direction of the crystal dislocation, and is known as the *Burgers’ vector*. The boundary between slipped and unslipped atoms on the slip plane is the dislocation line, shown as a dotted line. At position e , the dislocation line is perpendicular to the Burgers’ vector, so these two quantities lie in the slip plane. A dislocation so situated is called an *edge* dislocation, and is constrained to move only in the slip plane defined by the dislocation line and the Burgers’ vector.

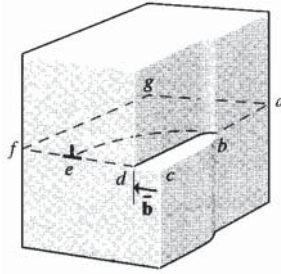


Figure 4: The mixed dislocation.

At position b , a spiral-like defect is formed such that a circular transit around the dislocation line ends on a plane a distance $\bar{b}$ from the starting point. Now the defect is known as a *screw* dislocation. The dislocation line is now parallel to the Burgers’ vector, so these two quantities do not define a unique slip plane the way an edge dislocation does. A screw dislocation can therefore *cross-slip* to another easy-glide plane passing through the dislocation line, and this mechanism enables screw dislocations to maneuver around obstacles that might otherwise impede their motion. Edge dislocations are more easily pinned, since they must “climb” by diffusion of vacancies to surmount obstacles as illustrated in Fig. 5.

As the curved dislocation line is traversed from point b to point e , the dislocation changes gradually from screw to edge character. At intermediate points the dislocation has both edge and screw character, and is known as a *mixed* dislocation.

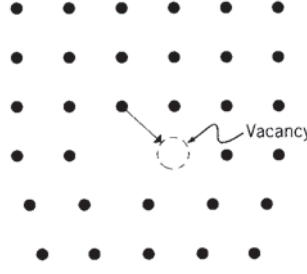


Figure 5: Dislocation climb by vacancy diffusion.

Dislocation-controlled yield

Single crystals tend to slip on their most closely packed planes, and in directions of minimum atomic separation distance. The distances between planes is maximum for the close-packed planes, so these are the most loosely bonded. Slip in close-packed directions minimizes the distance the stresses need to displace the slipping atoms. Both of these act to minimize the energy needed for slip. There are 12 such slip systems in the face-centered cubic (fcc) systems; using Miller indices, these are the $\{111\}$ planes and the $\langle 110 \rangle$ directions. There are 4 independent nonparallel (111) planes, and 3 independent $[110]$ directions in each plane.

Table 2: Critical Resolved shear stress for single crystals of various materials.

Material	crystal type	slip system	τ_{crss} , MPa
Nickel	fcc	$\{111\} \langle 110 \rangle$	5.7
Copper	fcc	$\{111\} \langle 110 \rangle$	0.98
Gold	fcc	$\{111\} \langle 110 \rangle$	0.90
Silver	fcc	$\{111\} \langle 110 \rangle$	0.60
Magnesium	hcp	$\{1101\} \langle 001 \rangle$	0.81
NaCl	cubic	$\{110\} \langle 110 \rangle$	0.75

Slip occurs when the shear stress on the slip plane, and in the slip direction, reaches a value τ_{crss} , the *critical resolved shear stress*; experimental values for τ_{crss} are listed in Table 2 for a number of single crystal materials. The resolved shear stress corresponding to an arbitrary stress state can be computed using the transformation relations of Module 10. In a simple tension test it can be written by inspection of Fig. 6 as

$$\tau_{rss} = \frac{P \cos \theta}{A_s} = \frac{P \cos \theta}{A_0 / \cos \phi} = \sigma (\cos \theta \cos \phi) \equiv \frac{\sigma}{m}$$

where m is a structure factor dependent on the orientation of the slip system relative to the applied tensile stress. For single crystals of arbitrary alignment, the yield stress will then be of the form

$$\sigma_Y = \tau_{crss} \cdot m \quad (1)$$

This is known as “Schmid’s Law,” and m is the “Schmid factor.”

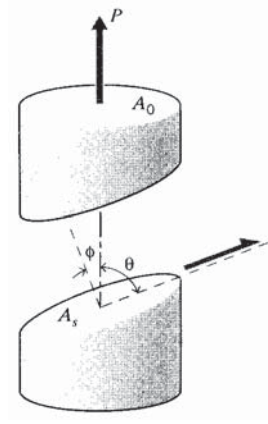


Figure 6: Critical resolved shear stress.

The yield stress will generally be higher in polycrystalline materials, since many of the grains will be oriented unfavorably (have high Schmid factors). Equation 1 can be modified for polycrystalline systems as

$$\sigma_Y = \tau_{crss} \cdot \bar{m}$$

where $\bar{m}$ is an equivalent Schmid factor that is generally somewhat higher than a simple average over all the individual grains; for fcc and bcc systems $\bar{m} \approx 3$.

Strain energy in dislocations

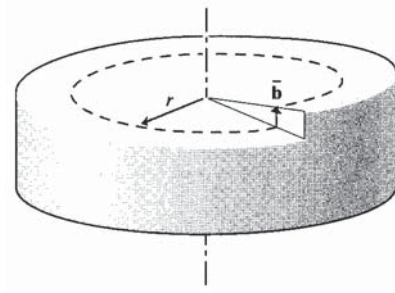


Figure 7: Shear strain associated with a screw dislocation.

Many calculations in dislocation mechanics are done more easily with energy concepts than with Newtonian force-displacement approaches. As seen in Fig. 7, the shear strain associated with a screw dislocation is the deflection $\bar{b}$ divided by the circumference of a circular path around the dislocation core:

$$\gamma = \frac{\bar{b}}{2\pi r} \quad (2)$$

where r is the distance from the dislocation core. Assuming Hookean elasticity, the corresponding strain energy per unit volume is

$$U = \int \tau d\gamma = \frac{1}{2} \tau \gamma = \frac{G\gamma^2}{2} = \frac{G\bar{b}^2}{8\pi^2 r^2}$$

The total strain energy associated with the screw dislocation is now obtained by integrating this over the volume around the dislocation:

$$U_{screw} = \int U dV = l \cdot \int_{r_0}^r \frac{G\bar{b}^2}{8\pi^2 r^2} 2\pi r dr$$

where here l is the length of the dislocation line and r_0 is the radius of the dislocation “core” inside which the energy is neglected. (Mathematically, the energy density increases without bound inside the core; however its volume becomes very small.) Taking $l = 1$ to obtain energy per unit length and carrying out the integration,

$$\boxed{U_{screw} = \frac{G\bar{b}^2}{4\pi} \ln \frac{r}{r_0} \approx G\bar{b}^2} \quad (3)$$

This last approximation should be read “scales as,” since it is arbitrary to select the limiting value r so that $\ln \frac{r}{r_0} \approx 4\pi$. The important conclusion is that the dislocation energy increases linearly with the shear modulus G and quadratically with the Burgers’ vector $\bar{b}$. A similar expression can be obtained for the strain energy per unit length of edge dislocation; it can be shown that

$$\boxed{U_{edge} = \frac{G\bar{b}^2}{1 - \nu}} \quad (4)$$

where ν is the Poisson’s ratio.

The dislocation energy represents an increase in the total energy of the system, which the material will try to eliminate if possible. For instance, two dislocations of opposite sign will be attracted to one another, since their strain fields will tend to cancel and lower the energy. Conversely, two dislocations of same sign will repel one another. The force of this attraction or repulsion will scale as

$$F dr = dU \Rightarrow F_{screw} \approx \frac{G\bar{b}^2}{r}$$

where here r is the distance between dislocations.

Dislocation motion and hardening

The ductility of crystalline materials is determined by dislocation mobility, and factors that impede dislocation motion can produce dramatic increases in the material’s yield strength. This increased resistance to plastic flow also raises the indentation hardness of the material, so strengthening of this sort is known as *hardening*. Alloying elements, grain boundaries, and even dislocations themselves can provide this impediment, and these provide the means by which the materials technologist controls yield. A thorough treatment of these important concepts must be left to subjects in physical metallurgy, but the following paragraphs will provide a brief introduction to some of them.

When one dislocation, moving on its slip plane under the influence of a driving shear stress, passes through another a “jog” will be created in the second dislocation as shown in Fig. 8.

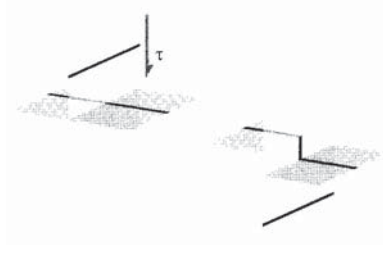


Figure 8: A dislocation jog.

The portion of the dislocation line in the jog is now no longer on its original glide plane, and is “pinned” in position. If the dislocation concentration is large, these jogs become a powerful impediment to plastic flow by dislocation motion. Paradoxically, the very dislocations that permit plastic flow in the first place can impede it if they become too numerous.



Figure 9: Dislocation bowing.

When a moving dislocation becomes pinned by jogs or other impediments, the shear stress τ that had been driving the dislocation now causes the line segment between the obstacles to bow forward as shown in Fig. 9, with an angle ϕ between adjacent segments. The extra length of the bowed line represents an increase in the strain energy of the dislocation, and if the shear stress were not present the line would straighten out to reduce this energy. The line acts similarly to an elastic band, with a “line tension” T that acts to return the line to a straight shortest-distance path between pinning points. The units of dislocation energy per unit length (N-m/m) are the same as simple tension, and we can write

$$T = \frac{\partial E}{\partial l} \approx Gb^2$$

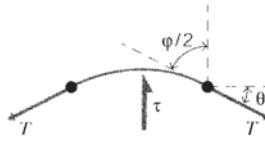


Figure 10: Force balance on dislocation segment.

As shown in Fig. 10, a free-body diagram of the line segment between two pinning points gives a force balance of the form

$$2T \sin \frac{d\theta}{2} = \tau \bar{b} \cdot r d\theta$$

where here r is the radius of curvature of the line (not the distance from the dislocation, as in Eqn. 2). Rearranging and canceling the $d\theta$ factor,

$$\tau = \frac{G\bar{b}}{r} \quad (5)$$

This relation gives the curvature of the dislocation in terms of the shear stress acting on it. The maximum shear stress is that needed to bend the dislocation into a semicircle (smallest r), after which the dislocation expands spontaneously. When the loops meet, annihilation occurs at that point, spawning a new dislocation line embedded in a circular loop. The process can be repeated with the new dislocation as well, and by this mechanism a large number of dislocations can be spawned as shown in Fig. 11. This is the “Frank-Read” source, and is an important means by which dislocations can multiply during plastic deformation. The increasing number of dislocations leads to more and more entanglements, with jogs acting as pinning points.



Figure 11: The Frank-Read dislocation source.

Equation 5 also provides an estimate of the influence of dislocation density on yield strength. If the obstacles pinning dislocation motion are “soft” the dislocation will be able to overcome them at relatively low driving stress, corresponding to a low critical angle ϕ_c . But as the obstacle becomes “harder,” i.e. provides more resistance to dislocation motion, the angle approaches zero and the radius of curvature becomes on the order of the obstacle spacing L . The shear stress needed to overcome such obstacles is then

$$\tau \approx \frac{G\bar{b}}{L}$$

When the hard obstacles arise from jogs created by intersections with other dislocations, the obstacle spacing L can be written in terms of the dislocation density. If the number of dislocations passing through a unit area is ρ , the number of dislocations encountered in moving along a straight line will be proportional to $\sqrt{\rho}$. The spacing between them is proportional to the reciprocal of this, so $\tau \propto G\bar{b}\sqrt{\rho}$. The yield stress is then the stress τ_0 needed to move dislocations in the absence of interfering dislocations, plus that needed to break through the obstacles; this can be written as

$$\boxed{\tau_Y = \tau_0 + AG\bar{b}\sqrt{\rho}} \quad (6)$$

where A is a constant that has been found to vary between 0.3 and 0.6 for a number of fcc, bcc and polycrystalline metals as well as some ionic crystals. Experimental corroboration of this relation is provided in Fig. 12.

The action of plastic flow therefore creates new dislocations by Frank-Read and other sources, which makes the material harder and harder, i.e. increasingly resistant to further plastic flow.

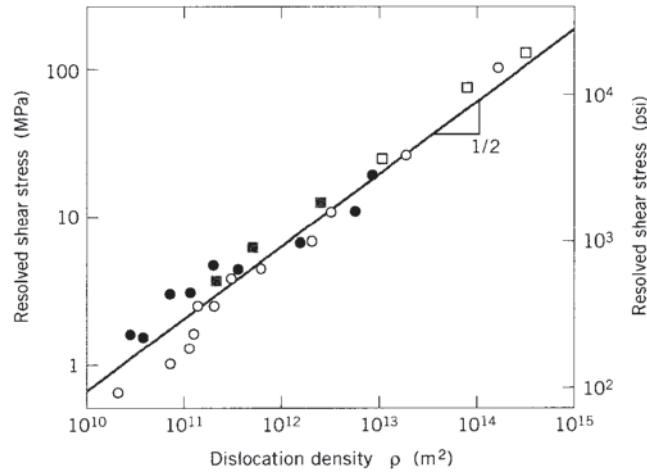


Figure 12: Effect of dislocation density ρ on critical resolved shear stress (for copper single and polycrystals, from T.H. Courtney, *Mechanical Behavior of Materials*, McGraw-Hill, 1990).

Eventually the yield stress for continued deformation becomes larger than the fracture stress, and the material will now break before it deforms further. If continued working of the material is desired, the number of dislocations must be reduced, for instance by thermal annealing. Annealing can produce *recovery* (dislocation climb around obstacles by vacancy diffusion) or recrystallization of new dislocation-free grains.

Grain boundaries act to impede dislocation motion, since the slip systems in adjoining grains will usually not line up; increases in yield strength arising from this mechanism are called *boundary strengthening*. Fine-grained metals have increased grain boundary area and thus have higher yield strengths than coarse-grained ones. The influence of grain size can often be described by the *Hall-Petch* formula

$$\sigma_Y = \sigma_0 + k_Y d^{-1/2} \quad (7)$$

where σ_0 is the lattice friction stress needed to move dislocations and K is a constant. This relation is essentially empirical, but it can be rationalized by viewing the second term as being related to the stress needed to activate a new mobile dislocation in the unfavorably oriented grain.

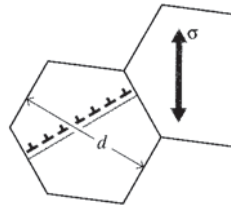


Figure 13: Dislocation pileup at a grain boundary.

As dislocations pile up against the boundary in the originally deforming grain, they act much like a crack whose length scales with the grain size d as shown in Fig. 13: the larger

the grain, the more dislocations in the pileup, the larger the virtual crack. Since the stress in front of a sharp crack of length a scales² as $\sqrt{a}$, the stress in front of the crack containing the dislocation pileup is increased by a factor that scales with $\sqrt{d}$. When this stress exceeds that needed to generate a new dislocation the unfavorably oriented grain begins to deform by dislocation motion. This stress diminishes according to $d^{-1/2}$ as the size of the original grain is scaled down, thus strengthening the metal according to the Hall-Petch relationship. Grain size is determined by the balance between nucleation and growth rates as the metal is solidified, and these are in turn controllable by the cooling rates imposed. This is an important example of processing-structure-property control available to the materials technologist.

A related phenomenon accounts for the very high strengths (≈ 4 GPa, or 600 kpsi) of piano wire, a eutectoid steel that has been drawn through a sequence of reducing dies to obtain a small final diameter. The “pearlitic” structure obtained on cooling this steel through the eutectoid temperature is a two-phase mixture of Fe_3C (“cementite”) in bcc iron (“ferrite”). As the diameter is reduced during drawing, the ferrite cells are reduced as well, forming a structure analogous to a fine-grained metal. The cell boundaries restrict dislocation motion, leading to the very high yield strengths.

Impurity atoms in solid solution can also serve to harden a crystalline material by impeding dislocation motion; this is called *solution strengthening*. An impurity atom smaller than the atoms of the host lattice will create an approximately spherical tensile field around itself which will attract the compressive regions around mobile dislocations, and a larger impurity atom will tend to trap the tensile region of nearby dislocations. On average, the population of dislocations will maneuver so as to lower their strain energies by associating with the nonuniform strain fields around impurities. This association impedes dislocation motion, which inhibits plastic flow and increases the yield stress.

Solution hardening is not usually an especially effective strengthening mechanism in commercial materials, largely because the solubility of impurity atoms is not sufficient to generate an appreciable number of obstacles. One important exception to this is the iron-carbon, or steel, system. If steel at approximately the eutectoid carbon composition (0.8% C) is cooled rapidly from above the eutectoid temperature of 723°C , the carbon atoms can become trapped in the iron lattice at much higher concentrations than bcc iron’s equilibrium carbon solubility would normally allow. (This tendency for trapping can be enhanced by alloying elements such as chromium and molybdenum, which have an affinity for carbon and thus reduce its ability to diffuse away.) To accommodate these metastable impurity atoms, the iron lattice transforms to a body-centered tetragonal form named *martensite* (see Fig. 14), with a strong nonspherical strain field around the carbon atoms. These tetragonal distortions are very effective impediments to dislocation motion, making martensite an extremely hard phase. The periodic water quenches a blacksmith uses during metalworking is done (perhaps without the smith knowing why it works) to tailor the material’s hardness by developing martensitic inclusions in the steel.

Martensite is so hard and brittle that the rapidly quenched steel must usually be *tempered* by heating it to approximately 400°C for an hour or so. This allows diffusion of carbon to take place, creating a dispersion of cementite inclusions; it also permits recovery of the dislocations present in the martensite. The resulting material is much tougher than the as-formed martensitic steel, but still retains a high strength level due to the strengthening effect of the carbide inclusions.

²See Module 16.

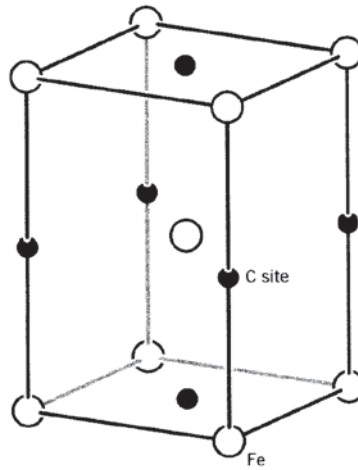


Figure 14: The body-centered tetragonal structure of martensite.

Kinetics of creep in crystalline materials

“Creep” is the term used to describe the tendency of many materials to exhibit continuing deformation even though the stress is held constant. Viscoelastic polymers exhibit creep, as was discussed in Module 19. However, creep also occurs in polycrystalline metallic and ceramic systems, most importantly when the the temperature is higher than approximately half their absolute melting temperature. This high-temperature creep can occur at stresses less than the yield stress, but is related to this module’s discussion of dislocation-controlled yield since dislocation motion often underlies the creep process as well.

High-temperature creep is of concern in such applications as jet engines or nuclear reactors. This form of creep often consists of three distinct regimes as seen in Fig. 15: *primary* creep, in which the material appears to harden so the creep rate diminishes with time; *secondary* or steady state creep, in which hardening and softening mechanisms appear to balance to produce a constant creep rate $\dot{\epsilon}_{II}$; and *tertiary* creep in which the material softens until creep rupture occurs. The entire creep curve reflects a competition between hardening mechanisms such as dislocation pileup, and mechanisms such as dislocation climb and cross-slip which are termed *recovery* and which augment dislocation mobility.

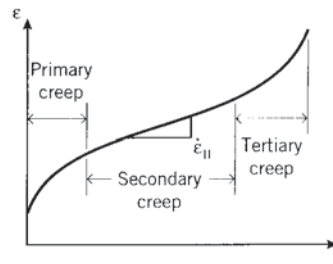


Figure 15: The three stages of creep.

In most applications the secondary regime consumes most of the time to failure, so much of

the modeling effort has been directed to this stage. The secondary creep rate $\dot{\epsilon}_{II}$ can often be described by a general nonlinear expression of the form

$$\boxed{\dot{\epsilon}_{II} = A\sigma^m \exp \frac{-E_c^*}{RT}} \quad (8)$$

where A and m are adjustable constants, E_c^* is an apparent activation energy for creep, σ is the stress, R is the Gas Constant (to be replaced by Boltzman's constant if a molar basis is not used) and T is the absolute temperature. This is known as the *Weertman-Dorn equation*.

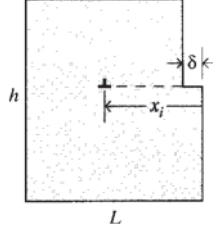


Figure 16: Dislocation motion and creep rate.

The plastic flow rate is related directly to dislocation velocity, which can be visualized by considering a section of material of height h and width L as shown in Fig. 16. A single dislocation, having traveled in the width direction for the full distance L will produce a transverse deformation of $\delta_i = \bar{b}$. If the dislocation has propagated through the crystal only a fraction x_i/L of the width, the deformation can be reduced by this same fraction: $\delta_i = \bar{b}(x_i/L)$. The total deformation in the crystal is then the sum of the deformations contributed by each dislocation:

$$\delta = \sum_i \delta_i = \sum_i \bar{b}(x_i/L)$$

The shear strain is the ratio of the transverse deformation to the height over which it is distributed:

$$\gamma = \frac{\delta}{h} = \frac{\bar{b}}{Lh} \sum_i x_i$$

The value $\sum_i x_i$ can be replaced by the quantity $N\bar{x}$, where N is the number of dislocations in the crystal segment and $\bar{x}$ is the average propagation distance. We can then write

$$\gamma = \rho \bar{b} \bar{x}$$

where $\rho = N/Lh$ is the dislocation density in the crystal. The shear strain *rate* $\dot{\gamma}$ is then obtained by differentiation:

$$\boxed{\dot{\gamma} = \rho \bar{b} v} \quad (9)$$

where $v = \dot{\bar{x}}$ is the average dislocation velocity. Hence the creep rate scales directly with the dislocation velocity.

To investigate the temperature and stress dependence of this velocity, we consider rate at which dislocations can overcome obstacles to be yet another example of a thermally activated, stress aided rate process and write an Eyring equation for the creep rate:

$$\dot{\epsilon} \propto v \propto \exp \frac{-(E_d^* - \sigma V^*)}{kT} - \exp \frac{-(E_d^* + \sigma V^*)}{kT}$$

where V^* is an apparent activation volume. The second term here indicates that the activation barrier for motion in the direction of stress is augmented by the stress, and diminished for motions in the opposite direction. When we discussed yielding the stress was sufficiently high that motion in the direction opposing flow could be neglected. Here we are interested in creep taking place at relatively low stresses and at high temperature, so that reverse flow can be appreciable. Factoring,

$$\dot{\epsilon} \propto \exp \frac{-E_d^*}{RT} \left(\exp \frac{+\sigma V^*}{RT} - \exp \frac{-\sigma V^*}{RT} \right)$$

Since $\sigma V^* \ll RT$, we can neglect quadratic and higher order terms in the series expansion $e^x = 1 + x + (x^2/2!) + (x^3/3!) + \dots$ to give

$$\dot{\epsilon} = A \left(\frac{\sigma V^*}{RT} \right) \exp \frac{-(E_d^*)}{RT}$$

If now we neglect the temperature dependence in the preexponential factor in comparison with the much stronger temperature dependence of the exponential itself, this model predicts a creep rate in agreement with the Weertman-Dorn equation with $m = 1$.

Creep by dislocation glide occurs over the full range of temperatures from absolute zero to the melting temperature, although the specific equation developed above contains approximations valid only at higher temperature. The stresses needed to drive dislocation glide are on the order of a tenth the theoretical shear strength of $G/10$. At lower stresses the creep rate is lower, and becomes limited by the rate at which dislocations can climb over obstacles by vacancy diffusion. This is hinted at in the similarity of the activation energies for creep and self diffusion as shown in Fig. 17. (Note that these values also correlate with the tightness of the bond energy functions, as discussed in Module 1; diffusion is impeded in more tightly-bonded lattices.) Vacancy diffusion is another stress-aided thermally activated rate process, again leading to models in agreement with the Weertman-Dorn equation.

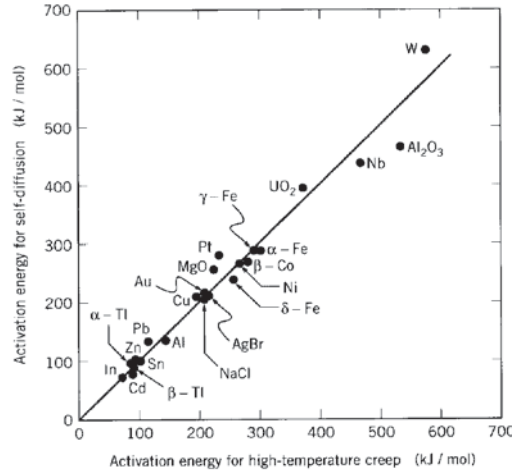


Figure 17: Correlation of activation energies for diffusion and creep.

Problems

1. The yield stresses σ_Y have been measured using steel and aluminum specimens of various grain sizes, as follows:

Material	d, μ	σ_Y , MPa
steel	60.5	160
	136	130
aluminum	11.1	235
	100	225

- (a) Determine the coefficients σ_0 and k_Y in the Hall-Petch relation (Eqn. 7) for these two materials.
- (b) Determine the yield stress in each material for a grain size of $d = 30\mu$.

Statistics of Fracture

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Introduction

One particularly troublesome aspect of fracture, especially in high-strength and brittle materials, is its *variability*. The designer must be able to cope with this, and limit stresses to those which reduce the *probability* of failure to an acceptably low level. Selection of an acceptable level of risk is a difficult design decision itself, obviously being as close to zero as possible in cases where human safety is involved but higher in doorknobs and other inexpensive items where failure is not too much more than a nuisance. The following sections will not replace a thorough study of statistics, but will introduce at least some of the basic aspects of statistical theory needed in design against fracture. The text by Collins¹ includes an extended treatment of statistical analysis of fracture and fatigue data, and is recommended for further reading.

Basic statistical measures

The value of tensile strength σ_f cited in materials property handbooks is usually the *arithmetic mean*, simply the sum of a number of individual strength measurements divided by the number of specimens tested:

$$\overline{\sigma_f} = \frac{1}{N} \sum_{i=1}^N \sigma_{f,i} \quad (1)$$

where the overline denotes the mean and $\sigma_{f,i}$ is the measured strength of the i^{th} (out of N) individual specimen. Of course, not all specimens have strengths exactly equal to the mean; some are weaker, some are stronger. There are several measures of how widely scattered is the distribution of strengths, one important one being the sample *standard deviation*, a sort of root mean square average of the individual deviations from the mean:

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (\overline{\sigma_f} - \sigma_{f,i})^2} \quad (2)$$

The significance of s to the designer is usually in relation to how large it is compared to the mean, so the *coefficient of variation*, or C.V., is commonly used:

¹Collins, J.A., *Failure of Materials in Mechanical Design*, Wiley, 1993.

$$\text{C.V.} = \frac{s}{\bar{\sigma}_f}$$

This is often expressed as a percentage. Coefficients of variation for tensile strength are commonly in the range of 1–10%, with values much over that indicating substantial inconsistency in the specimen preparation or experimental error.

Example 1

In order to illustrate the statistical methods to be outlined in this Module, we will use a sequence of thirty measurements of the room-temperature tensile strength of a graphite/epoxy composite². These data (in kpsi) are: 72.5, 73.8, 68.1, 77.9, 65.5, 73.23, 71.17, 79.92, 65.67, 74.28, 67.95, 82.84, 79.83, 80.52, 70.65, 72.85, 77.81, 72.29, 75.78, 67.03, 72.85, 77.81, 75.33, 71.75, 72.28, 79.08, 71.04, 67.84, 69.2, 71.53. Another thirty measurements from the same source, but taken at 93°C and -59°C, are given in Probs. 2 and 3, and can be subjected to the same treatments as homework.

There are several computer packages available for doing statistical calculations, and most of the procedures to be outlined here can be done with spreadsheets. The Microsoft Excel functions for mean and standard deviation are `average()` and `stdev()`, where the arguments are the range of cells containing the data. These give for the above data

$$\bar{\sigma}_f = 73.28, \quad s = 4.63 \text{ (kpsi)}$$

The coefficient of variation is $\text{C.V.} = (4.63/73.28) \times 100\% = 6.32\%$.

The normal distribution

A more complete picture of strength variability is obtained if the number of individual specimen strengths falling in a discrete strength interval $\Delta\sigma_f$ is plotted versus σ_f in a *histogram* as shown in Fig. 1; the maximum in the histogram will be near the mean strength and its width will be related to the standard deviation.

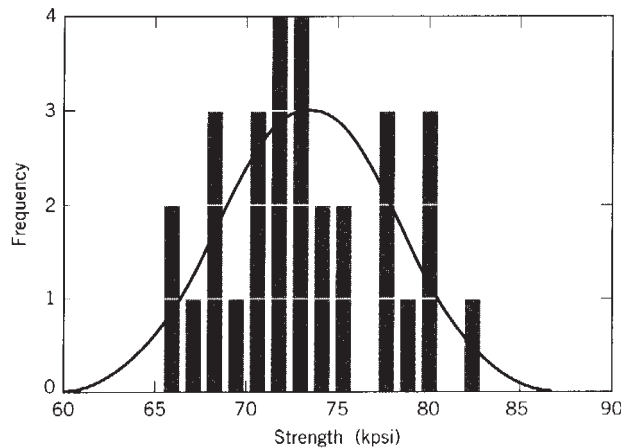


Figure 1: Histogram and normal distribution function for the strength data of Example 1.

²P. Shyprykevich, *ASTM STP 1003*, pp. 111–135, 1989.

As the number of specimens increases, the histogram can be drawn with increasingly finer $\Delta\sigma_f$ increments, eventually forming a smooth *probability distribution function*, or “pdf”. The mathematical form of this function is up to the material (and also the test method in some cases) to decide, but many phenomena in nature can be described satisfactorily by the *normal*, or *Gaussian*, function:

$$f(X) = \frac{1}{\sqrt{2\pi}} \exp \frac{-X^2}{2}, \quad X = \frac{\sigma_f - \bar{\sigma}_f}{s} \quad (3)$$

Here X is the *standard normal variable*, and is simply how many standard deviations an individual specimen strength is away from the mean. The factor $1/\sqrt{2\pi}$ normalizes the function so that its integral is unity, which is necessary if the specimen is to have a 100% chance of failing at *some* stress. In this expression we have assumed that the measure of standard deviation determined from Eqn. 2 based on a discrete number of specimens is acceptably close to the “true” value that would be obtained if every piece of material in the universe could somehow be tested.

The normal distribution function $f(X)$ plots as the “bell curve” familiar to all grade-conscious students. Its integral, known as the *cumulative distribution function* or $P_f(X)$, is also used commonly; its ordinate is the probability of fracture, also the fraction of specimens having a strength lower than the associated abscissal value. Since the normal pdf has been normalized, the cumulative function rises with an S-shaped or sigmoidal shape to approach unity at large values of X . The two functions $f(X)$ and $F(X)$ are plotted in Fig. 2, and tabulated in Tables 1 and 2 of the Appendix attached to this module. (Often the probability of survival $P_s = 1 - P_f$ is used as well; this curve begins at near unity and falls in a sigmoidal shape toward zero as the applied stress increases.)

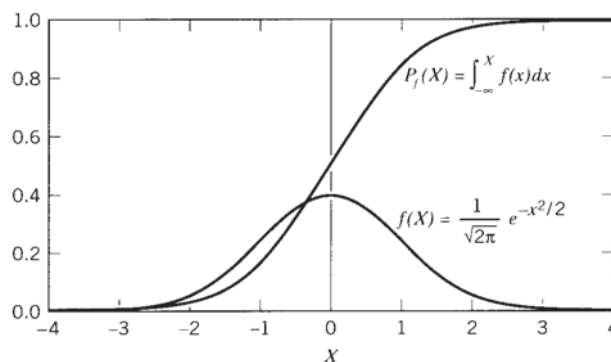


Figure 2: Differential $f(X)$ and cumulative $P_f(X)$ normal probability functions.

One convenient means of determining whether or not a particular set of measurements is normally distributed involves using special graph paper (a copy is included in the Appendix) whose ordinate has been distorted to make the sigmoidal cumulative distribution P_f plot as a straight line. (Sometimes it is easier to work with straight lines on curvy paper than curvy lines on straight paper.) Experimental data are ranked from lowest to highest, and each assigned a rank based on the fraction of strengths having higher values. If the ranks are assigned as $i/(N + 1)$, where i is the position of a datum in the ordered list and N is the number of specimens, the ranks are always greater than zero and less than one; this facilitates plotting.

The degree to which these rank-strength data plot as straight lines on normal probability paper is then a visual measure of how well the data are described by a normal distribution. The

best-fit straight line through the points passes the 50% cumulative fraction line at the sample mean, and its slope gives the standard distribution. Plotting several of these lines, for instance for different processing conditions of a given material, is a convenient way to characterize the strength differences arising from the two conditions (See Prob. 2).

Example 2

For our thirty-specimen test population, the ordered and ranked data are:

i	$\sigma_{f,i}$	$P_f = \frac{i}{N+1}$
1	65.50	0.0323
2	65.67	0.0645
3	67.03	0.0968
4	67.84	0.1290
5	67.95	0.1613
6	68.10	0.1935
7	69.20	0.2258
8	70.65	0.2581
9	71.04	0.2903
10	71.17	0.3226
11	71.53	0.3548
12	71.75	0.3871
13	72.28	0.4194
14	72.29	0.4516
15	72.5	0.4839
16	72.85	0.5161
17	72.85	0.5484
18	73.23	0.5806
19	73.80	0.6129
20	74.28	0.6452
21	75.33	0.6774
22	75.78	0.7097
23	77.81	0.7419
24	77.81	0.7742
25	77.90	0.8065
26	79.08	0.8387
27	79.83	0.8710
28	79.92	0.9032
29	80.52	0.9355
30	82.84	0.9677

When these are plotted using probability scaling on the ordinate, the graph in Fig. 3 is obtained.

The normal distribution function has been characterized thoroughly, and it is possible to infer a great deal of information from it for strength distributions that are close to normal. For instance, the cumulative normal distribution function (cdf) tabulated in Table 2 of the Appendix shows that that 68.3% of all members of a normal distribution lie within $\pm 1s$ of the mean, 95% lie within $\pm 1.96s$, and 99.865% lie within $\pm 3s$. It is common practice in much aircraft design to take $\overline{\sigma_f} - 3s$ as the safe fracture strength; then almost 99.9% of all specimens will have a strength at least this high. This doesn't really mean one out of every thousand airplane wings are unsafe; within the accuracy of the theory, 0.1% is a negligible number, and the $3s$

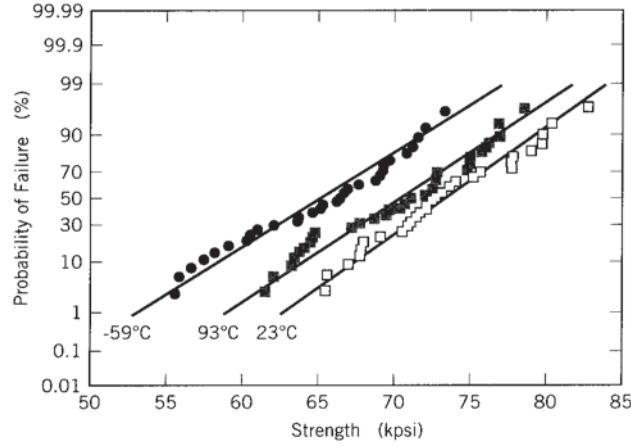


Figure 3: Probabilty plot of cumulative probability of failure for the strength data of Example 1. Also shown are test data taken at higher and lower temperature.

tolerance includes essentially the entire population³. Having to reduce the average strength by $3s$ in design can be a real penalty for advanced materials such as composites that have high strengths but also high variability due to their processing methods being relatively undeveloped. This is a major factor limiting the market share of these advanced materials.

Beyond the visual check of the linearity of the probability plot, several “goodness-of-fit” tests are available to assess the degree to which the population can reasonably be defined by the normal (or some other) distribution function. The “Chi-square” test is often used for this purpose. Here a test statistic measuring how far the observed data deviate from those expected from a normal distribution, or any other proposed distribution, is

$$\begin{aligned}\chi^2 &= \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ &= \sum_{i=1}^N \frac{(n_i - Np_i)^2}{Np_i}\end{aligned}$$

where n_i is the number of specimens actually failing in a strength increment $\Delta\sigma_{f,i}$, N is the total number of specimens, and p_i is the probability expected from the assumed distribution of a specimen having having a strength in that increment.

Example 3

To apply the Chi-square test to our 30-test population, we begin by counting the number of strengths falling in selected strength increments, much as if we were constructing a histogram. We choose five increments to obtain reasonable counts in each increment. The number *expected* to fall in each increment is determined from the normal pdf table, and the square of the difference calculated.

³ “Six-sigma” has become a popular goal in manufacturing, which means that only one part out of approximately a billion will fail to meet specification.

Lower Limit	Upper Limit	Observed Frequency	Expected Frequency	Chisquare
0	69.33	7	5.9	0.198
69.33	72.00	5	5.8	0.116
72.00	74.67	8	6.8	0.214
74.67	77.33	2	5.7	2.441
77.33	∞	8	5.7	0.909
				$\chi^2 = 3.878$

The number of degrees of freedom for this Chi-square test is 4; this is the number of increments less one, since we have the constraint that $n_1 + n_2 + n_3 + n_5 = 30$.

Interpolating in the Chi-Square Distribution Table (Table 3 in the Appendix), we find that a fraction 0.44 of normally distributed populations can be expected to have χ^2 statistics of up to 3.88. Hence, it seems reasonable that our population can be viewed as normally distributed.

Usually, we ask the question the other way around: is the computed χ^2 so large that only a small fraction — say 5% — of normally distributed populations would have χ^2 values that high or larger? If so, we would reject the hypothesis that our population is normally distributed.

From the χ^2 Table, we read that $\alpha = 0.05$ for $\chi^2 = 9.488$, where α is the fraction of the χ^2 population with values of χ^2 greater than 9.488. Equivalently, values of χ^2 above 9.488 would imply that there is less than a 5% chance that a population described by a normal distribution would have the computed χ^2 value. Our value of 3.878 is substantially less than this, and we are justified in claiming our data to be normally distributed.

Several governmental and voluntary standards-making organizations have worked to develop standardized procedures for generating statistically allowable property values for design of critical structures⁴. One such procedure defines the “B-allowable” strength as that level for which we have 95% confidence that 90% of all specimens will have at least that strength. (The use of *two* percentages here may be confusing. We mean that if we were to measure the strengths of 100 groups each containing 10 specimens, at least 95 of these groups would have at least 9 specimens whose strengths exceed the B-allowable.) In the event the normal distribution function is found to provide a suitable description of the population, the B-basis value can be computed from the mean and standard deviation using the formula

$$B = \overline{\sigma_f} - k_B s$$

where k_b is $n^{-1/2}$ times the 95th quantile of the “noncentral t-distribution;” this factor is tabulated, but can be approximated by the formula

$$k_b = 1.282 + \exp(0.958 - 0.520 \ln N + 3.19/N)$$

Example 4

In the case of the previous 30-test example, k_B is computed to be 1.78, so this is less conservative than the 3s guide. The B-basis value is then

$$B = 73.28 - (1.78)(4.632) = 65.05$$

Having a distribution function available lets us say something about the confidence we can have in how reliably we have measured the mean strength, based on a necessarily limited number

⁴ *Military Handbook 17B*, Army Materials Technology Laboratory, Part I, Vol. 1, 1987.

of individual strength tests. A famous and extremely useful result in mathematical statistics states that, if the mean of a distribution is measured N times, the *distribution of the means* will have its own standard deviation s_m that is related to the mean of the underlying distribution s and *the number of determinations*, N as

$$s_m = \frac{s}{\sqrt{N}} \quad (4)$$

This result can be used to establish *confidence limits*. Since 95% of all measurements of a normally distributed population lie within 1.96 standard deviations from the mean, the ratio $\pm 1.96s/\sqrt{N}$ is the range over which we can expect 95 out of 100 measurements of the mean to fall. So even in the presence of substantial variability we can obtain measures of mean strength to any desired level of confidence; we simply make more measurements to increase the value of N in the above relation. The “error bars” often seen in graphs of experimental data are not always labeled, and the reader must be somewhat cautious: they are *usually* standard deviations, but they may indicate maximum and minimum values, and occasionally they are 95% confidence limits. The significance of these three is obviously quite different.

Example 5

Equation 4 tells us that were we to repeat the 30-test sequence of the previous example over and over (obviously with new specimens each time), 95% of the measured sample means would lie within the interval

$$73.278 - \frac{(1.96)(4.632)}{\sqrt{30}}, \quad 73.278 + \frac{(1.96)(4.632)}{\sqrt{30}} = 71.62, 74.94$$

The t distribution

The “ t ” distribution, tabulated in Table 4 of the Appendix, is similar to the normal distribution, plotting as a bell-shaped curve centered on the mean. It has several useful applications to strength data. When there are few specimens in the sample, the t -distribution should be used in preference to the normal distribution in computing confidence limits. As seen in the table, the t -value for the 95th percentile and the 29 degrees of freedom of our 30-test sample in Example 3 is 2.045. (The number of degrees of freedom is one less than the total specimen count, since the sum of the number of specimens having each recorded strength is constrained to be the total number of specimens.) The 2.045 factor replaces 1.96 in this example, without much change in the computed confidence limits. As the number of specimens increases, the t -value approaches 1.96. For fewer specimens the factor deviates substantially from 1.96; it is 2.571 for $n = 5$ and 3.182 for $n = 3$.

The t distribution is also useful in deciding whether two test samplings indicate significant differences in the populations they are drawn from, or whether any difference in, say, the means of the two samplings can be ascribed to expected statistical variation in what are two essentially identical populations. For instance, Fig. 3 shows the cumulative failure probability for graphite-epoxy specimens tested at three different temperatures, and it appears that the mean strength is reduced somewhat by high temperatures and even more by low temperatures. But are these differences real, or merely statistical scatter?

This question can be answered by computing a value for t using the means and standard deviations of any two of the samples, according to the formula

$$t = \frac{|\bar{\sigma}_{f1} - \bar{\sigma}_{f2}|}{\sqrt{\frac{s_1^2}{n_1-1} + \frac{s_2^2}{n_2-1}}} \quad (5)$$

This statistic is known to have a t distribution if the deviations s_1 and s_2 are not too different. The mean and standard deviation of the -59°C data shown in Fig. 3 are 65.03 and 5.24 respectively. Using Eqn. 5 to compare the room-temperature (23°C) and -59°C data, the t -statistic is

$$t = \frac{(73.28 - 65.03)}{\sqrt{\frac{(4.63)^2}{29} + \frac{(5.24)^2}{29}}} = 6.354$$

From Table 4 in the Appendix, we now look up the value of t for 29 degrees of freedom corresponding to 95% (or some other value, as desired) of the population. We do this by scanning the column for $F(t) = 0.975$ rather than 0.95, since the t distribution is symmetric and another 0.025 fraction of the population lies beyond $t = -0.975$. The t value for 95% ($F(t) = 0.975$) and 29 degrees of freedom is 2.045.

This result means that were we to select repeatedly any two arbitrary 30-specimen samples from a single population, 95% of these selections would have t -statistics as computed with Eqn. 5 less than 2.045; only 5% would produce larger values of t . Since the 6.354 t -statistic for the -59°C and 23°C samplings is much greater than 2.045, we can conclude that it is very unlikely that the two sets of data are from the same population. Conversely, we conclude that the two datasets are in fact statistically independent, and that temperature has a statistically significant effect on the strength.

The Weibull distribution

Large specimens tend to have lower average strengths than small ones, simply because large ones are more likely to contain a flaw large enough to induce fracture at a given applied stress. This effect can be measured directly, for instance by plotting the strengths of fibers versus the fiber circumference as in Fig. 4. For similar reasons, brittle materials tend to have higher strengths when tested in flexure than in tension, since in flexure the stresses are concentrated in a smaller region near the outer surfaces.

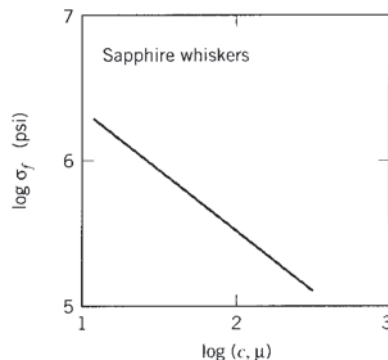


Figure 4: Effect of circumference c on fracture strength σ_f for sapphire whiskers. From L.J. Broutman and R.H. Krock, *Modern Composite Materials*, Addison-Wesley, 1967.

The hypothesis of the size effect led to substantial development effort in the statistical analysis community of the 1930's and 40's, with perhaps the most noted contribution being that of W. Weibull⁵ in 1939. Weibull postulated that the probability of survival at a stress σ , i.e. the probability that the specimen volume does not contain a flaw large enough to fail under the stress σ , could be written in the form

$$P_s(\sigma) = \exp \left[- \left(\frac{\sigma}{\sigma_0} \right)^m \right] \quad (6)$$

Weibull selected the form of this expression for its mathematical convenience rather than some fundamental understanding, but it has been found over many trials to describe fracture statistics well. The parameters σ_0 and m are adjustable constants; Fig. 5 shows the form of the Weibull function for two values of the parameter m . Materials with greater variability have smaller values of m ; steels have $m \approx 100$, while ceramics have $m \approx 10$. This parameter can be related to the coefficient of variation; to a reasonable approximation, $m \approx 1.2/\text{C.V.}$

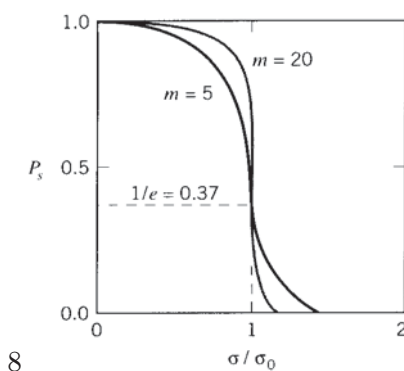


Figure 5: The Weibull function.

A variation on the normal probability paper graphical method outlined earlier can be developed by taking logarithms of Eqn. 6:

$$\ln P_s = - \left(\frac{\sigma}{\sigma_0} \right)^m$$

$$\ln(\ln P_s) = -m \ln \left(\frac{\sigma}{\sigma_0} \right)$$

Hence the double logarithm of the probability of exceeding a particular strength σ versus the logarithm of the strength should plot as a straight line with slope m .

Example 6

Again using the 30-test sample of the previous examples, an estimate of the σ_0 parameter can be obtained by plotting the survival probability (1 minus the rank) and noting the value of σ_f at which P_s drops to $1/e = 0.37$; this gives $\sigma_0 \approx 74$. (A more accurate regression method gives 75.46.) A tabulation of the double logarithm of P_s against the logarithm of σ_f/σ_0 is then

⁵See B. Epstein, *J. Appl. Phys.*, Vol. 19, p. 140, 1948 for a useful review of the statistical treatment of the size effect in fracture, and for a summary of extreme-value statistics as applied to fracture problems.

i	$\sigma_{f,i}$	$\ln \ln(1/P_s)$	$\ln(\sigma_{f,i}/\sigma_0)$
1	65.50	-3.4176	-0.1416
2	65.67	-2.7077	-0.1390
3	67.03	-2.2849	-0.1185
4	67.84	-1.9794	-0.1065
5	67.95	-1.7379	-0.1048
6	68.10	-1.5366	-0.1026
7	69.20	-1.3628	-0.0866
8	70.65	-1.2090	-0.0659
9	71.04	-1.0702	-0.0604
10	71.17	-0.9430	-0.0585
11	71.53	-0.8250	-0.0535
12	71.75	-0.7143	-0.0504
13	72.28	-0.6095	-0.0431
14	72.29	-0.5095	-0.0429
15	72.50	-0.4134	-0.0400
16	72.85	-0.3203	-0.0352
17	72.85	-0.2295	-0.0352
18	73.23	-0.1404	-0.0300
19	73.80	-0.0523	-0.0222
20	74.28	0.0355	-0.0158
21	75.33	0.1235	-0.0017
22	75.78	0.2125	0.0042
23	77.81	0.3035	0.0307
24	77.81	0.3975	0.0307
25	77.90	0.4961	0.0318
26	79.08	0.6013	0.0469
27	79.83	0.7167	0.0563
28	79.92	0.8482	0.0574
29	80.52	1.0083	0.0649
30	82.84	1.2337	0.0933

The Weibull plot of these data is shown in Fig. 6; the regression slope is 17.4.

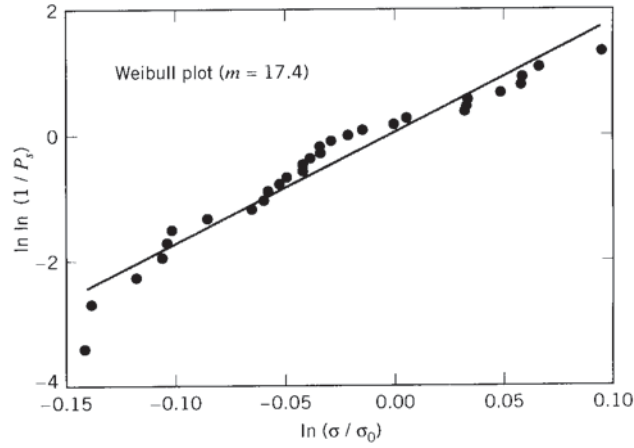


Figure 6: Weibull plot.

Similarly to the B-basis design allowable for the normal distribution, the B-allowable can also be computed from the Weibull parameters m and σ_0 . The procedure is⁶:

$$B = Q \exp \left[\frac{-V}{m\sqrt{N}} \right]$$

where Q and V are

$$Q = \sigma_0 (0.10536)^{1/m}$$

$$V = 3.803 + \exp \left[1.79 - 0.516 \ln(N) + \frac{5.1}{N} \right]$$

Example 7

The B-allowable is computed for the 30-test population as

$$Q = 75.46 (0.10536)^{1/17.4m} = 66.30$$

$$V = 3.803 + \exp \left[1.79 - 0.516 \ln(30) + \frac{5.1}{30} \right] = 5.03$$

$$B = 66.30 \exp \left[\frac{-5.03}{17.4\sqrt{30}} \right] = 62.89$$

This value is somewhat lower than the 65.05 obtained as the normal-distribution B-allowable, so in this case the Weibull method is a bit more lenient.

The Weibull equation can be used to predict the magnitude of the size effect. If for instance we take a reference volume V_0 and express the volume of an arbitrary specimen as $V = nV_0$, then the probability of failure at volume V is found by multiplying $P_s(V)$ by itself n times:

$$P_s(V) = [P_s(V_0)]^n = [P_s(V_0)]^{V/V_0}$$

$$\boxed{P_s(V) = \exp - \frac{V}{V_0} \left(\frac{\sigma}{\sigma_0} \right)^m} \quad (7)$$

Hence the probability of failure increases exponentially with the specimen volume. This is another danger in simple scaling, beyond the area vs. volume argument we outlined in Module 1.

Example 8

Solving Eqn. 7, the stress for a given probability of survival is

$$\sigma = \left[\frac{-\ln(P_s)}{(V/V_0)} \right]^{\frac{1}{n}} \sigma_0$$

Using $\sigma_0 = 75.46$ and $n = 17.4$ for the 30-specimen population, the stress for $P_s = .5$ and $V/V_0 = 1$ is $\sigma = 73.9$ kpsi. If now the specimen size is doubled, so that $V/V_0 = 2$, the probability of survival at this stress as given by Eqn. 7 drops to $P_s = 0.25$. If on the other hand the specimen size is halved ($V/V_0 = 0.5$), the probability of survival rises to $P_s = 0.71$.

⁶S.W. Rust, et al., *ASTM STP 1003*, p. 136, 1989. (Also Military Handbook 17.)

A final note of caution, a bit along the lines of the famous Mark Twain aphorism about there being “lies, damned lies, and statistics:” it is often true that populations of simple tensile or other laboratory specimens can be well described by classical statistical distributions. This should not be taken to imply that more complicated structures such as bridges and airplanes can be so neatly described. For instance, one aircraft study cited by Gordon⁷ found failures to occur randomly and uniformly over a wide range extending both above and below the statistically-based design safe load. Any real design, especially for structures that put human life at risk, must be checked in every reasonable way the engineer can imagine. This will include proof testing to failure, consideration of the worst possible environmental factors, consideration of construction errors resulting from difficult-to-manufacture designs, and so on almost without limit. Experience, caution and common sense will usually be at least as important as elaborate numerical calculations.

Problems

1. Ten strength measurements have produced a mean tensile strength of $\overline{\sigma_f} = 100$ MPa, with 95% confidence limits of ± 8 MPa. How many additional measurements would be necessary to reduce the confidence limits by half, assuming the mean and standard deviation of the measurements remains unchanged?
2. The thirty measurements of the tensile strength of graphite/epoxy composite listed in Example 1 were made at room temperature. Thirty additional tests conducted at 93°C gave the values (in kpsi): 63.40, 69.70, 72.80, 63.60, 71.20, 72.07, 76.97, 70.94, 76.22, 64.65, 62.08, 61.53, 70.53, 72.88, 74.90, 78.61, 68.72, 72.87, 64.49, 75.12, 67.80, 72.68, 75.09, 67.23, 64.80, 75.84, 63.87, 72.46, 69.54, 76.97. For these data:
 - (a) Determine the arithmetic mean, standard deviation, and coefficient of variation.
 - (b) Determine the 95% confidence limits on the mean strength.
 - (c) Determine whether the average strengths at 23°C and 93°C are statistically different.
 - (d) Determine the normal and Weibull B-allowable strengths.
 - (e) Plot the cumulative probability of failure P_f vs. the failure stress on normal probability paper.
 - (f) Do the data appear to be distributed normally, based on the χ^2 test?
 - (g) Plot the cumulative probability of survival P_s vs. the failure stress on Weibull probability paper.
 - (h) Determine the Weibull parameters σ_0 and m .
 - (i) Estimate how the mean strength would change if the specimens were made ten times smaller, or ten times larger.
3. Repeat the previous problem, but using data for -59°C: 55.62, 55.91, 56.73, 57.54, 58.28, 59.23, 60.39, 60.62, 61.1, 62.1, 63.69, 63.8, 64.7, 65.2, 65.33, 66.39, 66.43, 66.72, 67.05, 67.76, 68.84, 69.15, 69.3, 69.37, 69.82, 70.94, 71.39, 71.74, 72.2, 73.46.

⁷J.E. Gordon, *Structures*, Plenum Press, 1978.

Appendix - Statistical Tables and Paper

Following are several standard tables and graph papers that can be used in performing statistical calculations, in the event suitable software is not available.

1. Normal Distribution Table
2. Cumulative Normal Distribution Table
3. Chi-Square Table
4. t -Distribution
5. Normal Probability Paper
6. Weibull Paper

1. Normal Distribution

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3989	0.3989	0.3988	0.3986	0.3984	0.3982	0.3980	0.3977	0.3973
0.1	0.3970	0.3965	0.3961	0.3956	0.3951	0.3945	0.3939	0.3932	0.3925	0.3918
0.2	0.3910	0.3902	0.3894	0.3885	0.3876	0.3867	0.3857	0.3847	0.3836	0.3825
0.3	0.3814	0.3802	0.3790	0.3778	0.3765	0.3752	0.3739	0.3725	0.3712	0.3697
0.4	0.3683	0.3668	0.3653	0.3637	0.3621	0.3605	0.3589	0.3572	0.3555	0.3538
0.5	0.3521	0.3503	0.3485	0.3467	0.3448	0.3429	0.3410	0.3391	0.3372	0.3352
0.6	0.3332	0.3312	0.3292	0.3271	0.3251	0.3230	0.3209	0.3187	0.3166	0.3144
0.7	0.3123	0.3101	0.3079	0.3056	0.3034	0.3011	0.2989	0.2966	0.2943	0.2920
0.8	0.2897	0.2874	0.2850	0.2827	0.2803	0.2780	0.2756	0.2732	0.2709	0.2685
0.9	0.2661	0.2637	0.2613	0.2589	0.2565	0.2541	0.2516	0.2492	0.2468	0.2444
1.0	0.2420	0.2396	0.2371	0.2347	0.2323	0.2299	0.2275	0.2251	0.2227	0.2203
1.1	0.2179	0.2155	0.2131	0.2107	0.2083	0.2059	0.2036	0.2012	0.1989	0.1965
1.2	0.1942	0.1919	0.1895	0.1872	0.1849	0.1826	0.1804	0.1781	0.1758	0.1736
1.3	0.1714	0.1691	0.1669	0.1647	0.1626	0.1604	0.1582	0.1561	0.1539	0.1518
0.4	0.1497	0.1476	0.1456	0.1435	0.1415	0.1394	0.1374	0.1354	0.1334	0.1315
1.5	0.1295	0.1276	0.1257	0.1238	0.1219	0.1200	0.1182	0.1163	0.1145	0.1127
1.6	0.1109	0.1092	0.1074	0.1057	0.1040	0.1023	0.1006	0.0989	0.0973	0.0957
1.7	0.0940	0.0925	0.0909	0.0893	0.0878	0.0863	0.0848	0.0833	0.0818	0.0804
1.8	0.0790	0.0775	0.0761	0.0748	0.0734	0.0721	0.0707	0.0694	0.0681	0.0669
1.9	0.0656	0.0644	0.0632	0.0620	0.0608	0.0596	0.0584	0.0573	0.0562	0.0551
2.0	0.0540	0.1529	0.0519	0.0508	0.0498	0.0488	0.0478	0.0468	0.0459	0.0449
2.1	0.0440	0.0431	0.0422	0.0413	0.0404	0.0396	0.0387	0.0379	0.0371	0.0363
2.2	0.0355	0.0347	0.0339	0.0332	0.0325	0.0317	0.0310	0.0303	0.0297	0.0290
2.3	0.0283	0.0277	0.0270	0.0264	0.0258	0.0252	0.0246	0.0241	0.0235	0.0229
2.4	0.0224	0.0219	0.0213	0.0208	0.0203	0.0198	0.0194	0.0189	0.0184	0.0180
2.5	0.0175	0.0171	0.0167	0.0163	0.0158	0.0154	0.0151	0.0147	0.0143	0.0139
2.6	0.0136	0.0132	0.0129	0.0126	0.0122	0.0119	0.0116	0.0113	0.0110	0.0107
2.7	0.0104	0.0101	0.0099	0.0096	0.0093	0.0091	0.0088	0.0086	0.0084	0.0081
2.8	0.0079	0.0077	0.0075	0.0073	0.0071	0.0069	0.0067	0.0065	0.0063	0.0061
2.9	0.0060	0.0058	0.0056	0.0055	0.0053	0.0051	0.0050	0.0048	0.0047	0.0046
3.0	0.0044	0.0043	0.0042	0.0040	0.0039	0.0038	0.0037	0.0036	0.0035	0.0034
3.1	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026	0.0025	0.0025
3.2	0.0024	0.0023	0.0022	0.0022	0.0021	0.0020	0.0020	0.0019	0.0018	0.0018
3.3	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014	0.0013	0.0013
3.4	0.0012	0.0012	0.0012	0.0011	0.0011	0.0010	0.0010	0.0010	0.0009	0.0009
3.5	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007	0.0007	0.0007	0.0006
3.6	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005	0.0004
3.7	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003	0.0003	0.0003	0.0003
3.8	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002	0.0002	0.0002	0.0002	0.0002
3.9	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0001	0.0001

2. Cumulative Normal Distribution

[illegible]

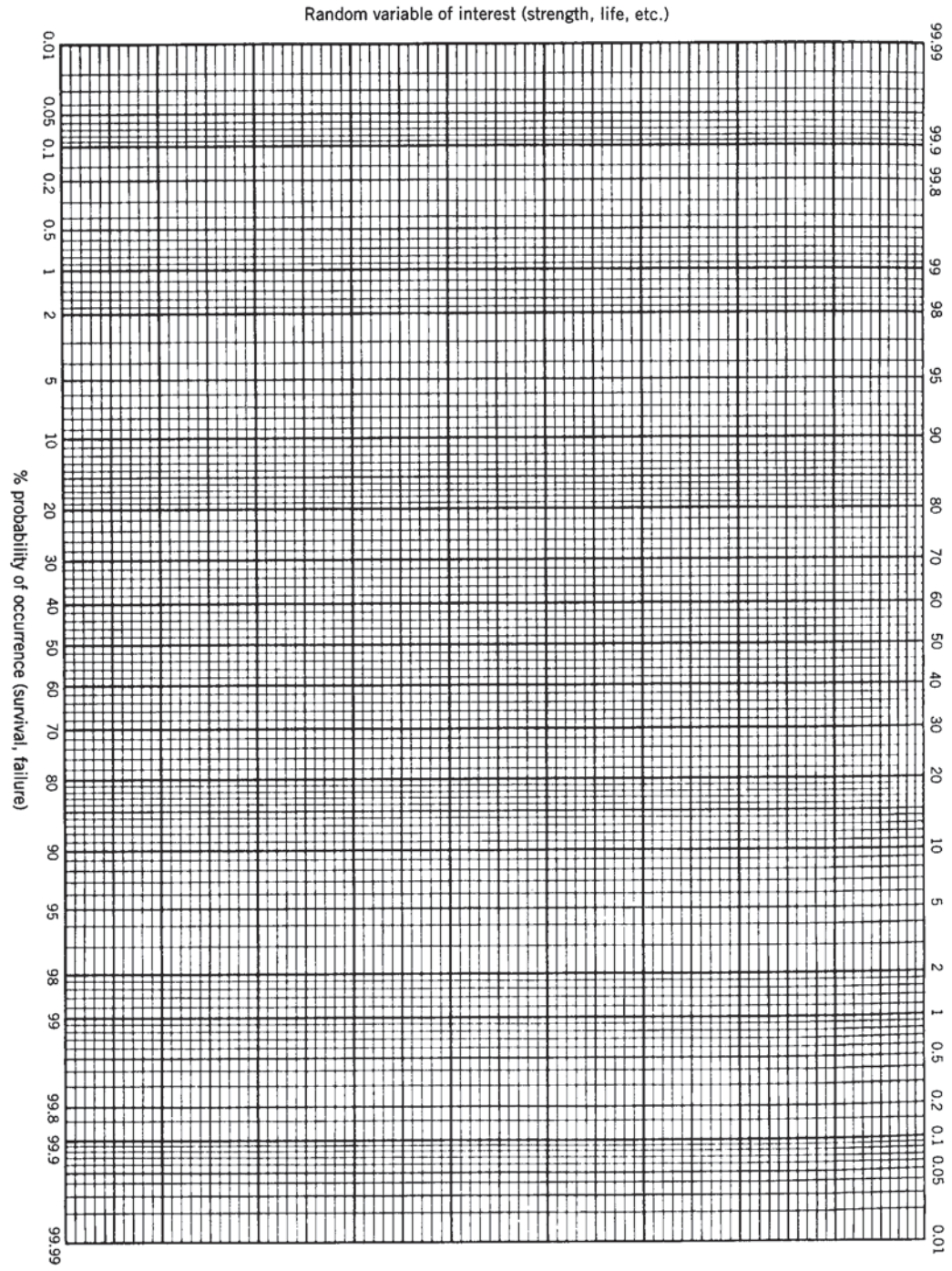
Maximum χ^2 values for $\alpha =$

Degrees of Freedom	0.995	0.990	0.975	0.950	0.900	0.750	0.500	0.250	0.100	0.050	0.025	0.010	0.005
1	0.0000393	0.000157	0.000982	0.00393	0.0158	0.102	0.455	1.32	2.71	3.84	5.02	6.63	7.88
2	0.0100	0.0201	0.0506	0.103	0.211	0.575	1.390	2.77	4.61	5.99	7.38	9.21	10.6
3	0.0717	0.115	0.216	0.352	0.584	1.21	2.37	4.11	6.25	7.81	9.35	11.3	12.8
4	0.207	0.297	0.484	0.711	1.06	1.92	3.36	5.39	7.78	9.49	11.1	13.3	14.9
5	0.412	0.554	0.831	1.15	1.61	2.67	4.35	6.63	9.24	11.1	12.8	15.1	16.7
6	0.676	0.872	1.24	1.64	2.20	3.45	5.35	7.84	10.6	12.6	14.4	16.8	18.5
7	0.989	1.24	1.69	2.17	2.83	4.25	6.35	9.04	12.0	14.1	16.0	18.5	20.3
8	1.34	1.65	2.18	2.73	3.49	5.07	7.34	10.2	13.4	15.5	17.5	20.1	22.0
9	1.73	2.09	2.70	3.33	4.17	5.90	8.34	11.4	14.7	16.9	19.0	21.7	23.6
10	2.16	2.56	3.25	3.94	4.87	6.74	9.34	12.5	16.0	18.3	20.5	23.2	25.2
11	2.60	3.05	3.82	4.57	5.58	7.58	10.3	13.7	17.3	19.7	21.9	24.7	26.8
12	3.07	3.57	4.40	5.23	6.30	8.44	11.3	14.8	18.5	21.0	23.3	26.2	28.3
13	3.57	4.11	5.01	5.89	7.04	9.30	12.3	16.0	19.8	22.4	24.7	27.7	29.8
14	4.07	4.66	5.63	6.57	7.79	10.2	13.3	17.1	21.1	23.7	26.1	29.1	31.3
15	4.60	5.23	6.26	7.26	8.55	11.0	14.3	18.2	22.3	25.0	27.5	30.6	32.8
16	5.14	5.81	6.91	7.96	9.31	11.9	15.3	19.4	23.5	26.3	28.8	32.0	34.3
17	5.70	6.41	7.56	8.67	10.1	12.8	16.3	20.5	24.8	27.6	30.2	33.4	35.7
18	6.26	7.01	8.23	9.39	10.9	13.7	17.3	21.6	26.0	28.9	31.5	34.8	37.2
19	6.84	7.63	8.91	10.1	11.7	14.6	18.3	22.7	27.2	30.1	32.9	36.2	38.6
20	7.43	8.26	9.59	10.9	12.4	15.5	19.3	23.8	28.4	31.4	34.2	37.6	40.0
21	8.03	8.90	10.3	11.6	13.2	16.3	20.3	24.9	29.6	32.7	35.5	38.9	41.4
22	8.64	9.54	11.0	12.3	14.0	17.2	21.3	26.0	30.8	33.9	36.8	40.3	42.8
23	9.26	10.20	11.7	13.1	14.8	18.1	22.3	27.1	32.0	35.2	38.1	41.6	44.2
24	9.89	10.90	12.4	13.8	15.7	19.0	23.3	28.2	33.2	36.4	39.4	43.0	45.6
25	10.50	11.50	13.1	14.6	16.5	19.9	24.3	29.3	34.4	37.7	40.6	44.3	46.9
26	11.20	12.20	13.8	15.4	17.3	20.8	25.3	30.4	35.6	38.9	41.9	45.6	48.3
27	11.80	12.90	14.6	16.2	18.1	21.7	26.3	31.5	36.7	40.1	43.2	47.0	49.6
28	12.50	13.60	15.3	16.9	18.9	22.7	27.3	32.6	37.9	41.3	44.5	48.3	51.0
29	13.10	14.30	16.0	17.7	19.8	23.6	28.3	33.7	39.1	42.6	45.7	49.6	52.3
30	13.80	15.00	16.8	18.5	20.6	24.5	29.3	34.8	40.3	43.8	47.0	50.9	53.7

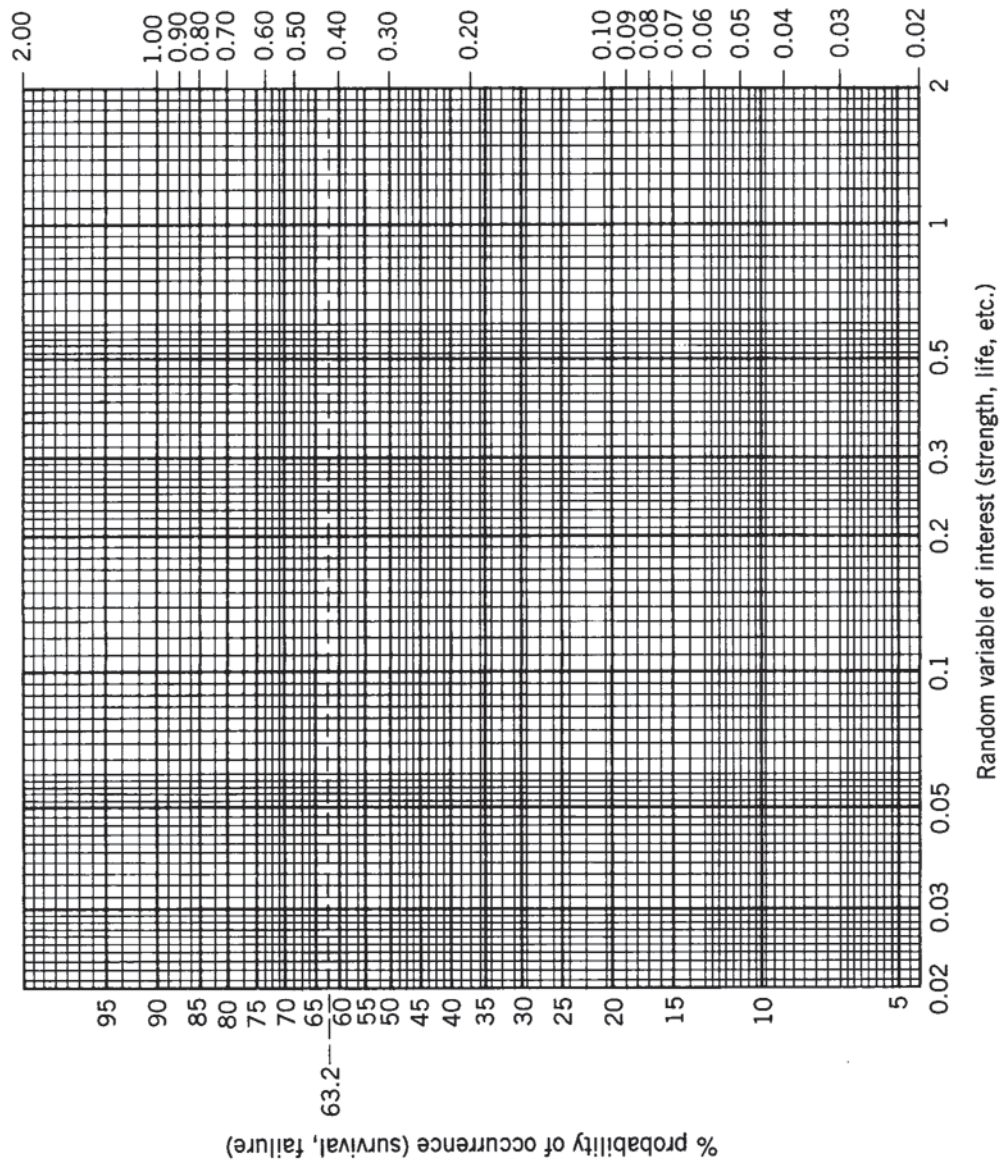
t-Distribution**Percentile****Degrees of
Freedom**

	50	80	90	95	98	99	99.90
1	1.000	3.078	6.314	12.706	31.821	63.657	636.610
2	0.816	1.886	2.920	4.303	6.965	9.925	31.598
3	0.765	1.638	2.353	3.182	4.541	5.841	12.941
4	0.741	1.533	2.132	2.776	3.747	4.604	8.610
5	0.727	1.476	2.015	2.571	3.365	4.032	6.859
6	0.718	1.440	1.943	2.447	3.143	3.707	5.959
7	0.711	1.415	1.895	2.365	2.998	3.499	5.405
8	0.706	1.397	1.860	2.306	2.896	3.355	5.041
9	0.703	1.383	1.833	2.262	2.821	3.250	4.781
10	0.700	1.372	1.812	2.228	2.764	3.169	4.587
11	0.697	1.363	1.796	2.201	2.718	3.106	4.437
12	0.695	1.356	1.782	2.179	2.681	3.055	4.318
13	0.694	1.350	1.771	2.160	2.650	3.012	4.221
14	0.692	1.345	1.761	2.145	2.624	2.977	4.140
15	0.691	1.341	1.753	2.131	2.602	2.947	4.073
16	0.690	1.337	1.746	2.120	2.583	2.921	4.015
17	0.689	1.333	1.740	2.110	2.567	2.898	3.965
18	0.688	1.330	1.734	2.101	2.552	2.878	3.922
19	0.688	1.328	1.729	2.093	2.539	2.861	3.883
20	0.687	1.325	1.725	2.086	2.528	2.845	3.850
21	0.686	1.323	1.721	2.080	2.518	2.831	3.819
22	0.686	1.321	1.717	2.074	2.508	2.819	3.792
23	0.685	1.319	1.714	2.069	2.500	2.807	3.767
24	0.685	1.318	1.711	2.064	2.492	2.797	3.745
25	0.684	1.316	1.708	2.060	2.485	2.787	3.725
26	0.684	1.315	1.706	2.056	2.479	2.779	3.707
27	0.684	1.314	1.703	2.052	2.473	2.771	3.690
28	0.683	1.313	1.701	2.048	2.467	2.763	3.674
29	0.683	1.311	1.699	2.045	2.462	2.756	3.659
30	0.683	1.310	1.697	2.042	2.457	2.750	3.646
40	0.681	1.303	1.684	2.021	2.423	2.704	3.551
60	0.679	1.296	1.671	2.000	2.390	2.660	3.460
120	0.677	1.289	1.658	1.980	2.358	2.617	3.373
infinity	0.674	1.282	1.645	1.960	2.326	2.576	3.291

5. Normal Probability Paper



6. Weibull Paper



Introduction to Fracture Mechanics

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Introduction

In 1983, the National Bureau of Standards (now the National Institute for Science and Technology) and Battelle Memorial Institute¹ estimated the costs for failure due to fracture to be \$119 billion per year in 1982 dollars. The dollars are important, but the cost of many failures in human life and injury is infinitely more so.

Failures have occurred for many reasons, including uncertainties in the loading or environment, defects in the materials, inadequacies in design, and deficiencies in construction or maintenance. Design against fracture has a technology of its own, and this is a very active area of current research. This module will provide an introduction to an important aspect of this field, since without an understanding of fracture the methods in stress analysis discussed previously would be of little use. We will focus on fractures due to simple tensile overstress, but the designer is cautioned again about the need to consider absolutely as many factors as possible that might lead to failure, especially when life is at risk.

The Module on the Dislocation Basis of Yield (Module 21) shows how the strength of structural metals – particularly steel – can be increased to very high levels by manipulating the microstructure so as to inhibit dislocation motion. Unfortunately, this renders the material increasingly brittle, so that cracks can form and propagate catastrophically with very little warning. An unfortunate number of engineering disasters are related directly to this phenomenon, and engineers involved in structural design must be aware of the procedures now available to safeguard against brittle fracture.

The central difficulty in designing against fracture in high-strength materials is that the presence of cracks can modify the local stresses to such an extent that the elastic stress analyses done so carefully by the designers are insufficient. When a crack reaches a certain critical length, it can propagate catastrophically through the structure, *even though the gross stress is much less than would normally cause yield or failure in a tensile specimen*. The term “fracture mechanics” refers to a vital specialization within solid mechanics in which the presence of a crack is assumed, and we wish to find quantitative relations between the crack length, the material’s inherent resistance to crack growth, and the stress at which the crack propagates at high speed to cause structural failure.

¹R.P. Reed et al., NBS Special Publication 647-1, Washington, 1983.

The energy-balance approach

When A.A. Griffith (1893–1963) began his pioneering studies of fracture in glass in the years just prior to 1920, he was aware of Inglis’ work in calculating the stress concentrations around elliptical holes², and naturally considered how it might be used in developing a fundamental approach to predicting fracture strengths. However, the Inglis solution poses a mathematical difficulty: in the limit of a perfectly sharp crack, the stresses approach infinity at the crack tip. This is obviously nonphysical (actually the material generally undergoes some local yielding to blunt the cracktip), and using such a result would predict that materials would have near-*zero* strength: even for very small applied loads, the stresses near crack tips would become infinite, and the bonds there would rupture. Rather than focusing on the crack-tip stresses directly, Griffith employed an energy-balance approach that has become one of the most famous developments in materials science³.

The strain energy per unit volume of stressed material is

$$U^* = \frac{1}{V} \int f dx = \int \frac{f}{A} \frac{dx}{L} = \int \sigma d\epsilon$$

If the material is linear ($\sigma = E\epsilon$), then the strain energy per unit volume is

$$U^* = \frac{E\epsilon^2}{2} = \frac{\sigma^2}{2E}$$

When a crack has grown into a solid to a depth a , a region of material adjacent to the free surfaces is unloaded, and its strain energy released. Using the Inglis solution, Griffith was able to compute just how much energy this is.

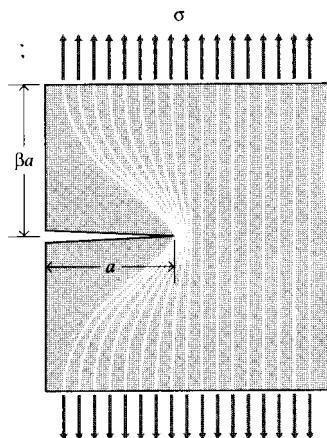


Figure 1: Idealization of unloaded region near crack flanks.

A simple way of visualizing this energy release, illustrated in Fig. 1, is to regard two triangular regions near the crack flanks, of width a and height βa , as being completely unloaded, while the remaining material continues to feel the full stress σ . The parameter β can be selected so as to

²See Module 16.

³A.A. Griffith, *Philosophical Transactions*, Series A, Vol. 221, pp. 163–198, 1920. The importance of Griffith’s work in fracture was largely unrecognized until the 1950’s. See J.E. Gordon, *The Science of Structures and Materials*, Scientific American Library, 1988, for a personal account of the Griffith story.

agree with the Inglis solution, and it turns out that for plane stress loading $\beta = \pi$. The total strain energy U released is then the strain energy per unit volume times the volume in both triangular regions:

$$U = -\frac{\sigma^2}{2E} \cdot \pi a^2$$

Here the dimension normal to the x - y plane is taken to be unity, so U is the strain energy released per unit thickness of specimen. This strain energy is *liberated* by crack growth. But in forming the crack, bonds must be broken, and the requisite bond energy is in effect *absorbed* by the material. The surface energy S associated with a crack of length a (and unit depth) is:

$$S = 2\gamma a$$

where γ is the surface energy (e.g., Joules/meter²) and the factor 2 is needed since two free surfaces have been formed. As shown in Fig. 2, the total energy associated with the crack is then the sum of the (positive) energy absorbed to create the new surfaces, plus the (negative) strain energy liberated by allowing the regions near the crack flanks to become unloaded.

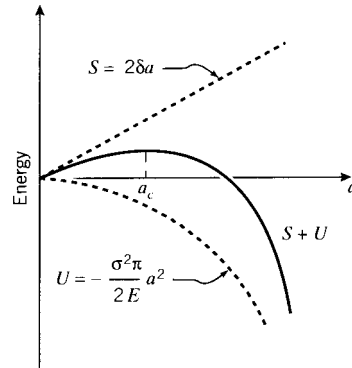


Figure 2: The fracture energy balance.

As the crack grows longer (a increases), the quadratic dependence of strain energy on a eventually dominates the surface energy, and beyond a critical crack length a_c the system can lower its energy by letting the crack grow still longer. Up to the point where $a = a_c$, the crack will grow only if the stress is increased. Beyond that point, crack growth is spontaneous and catastrophic.

The value of the critical crack length can be found by setting the derivative of the total energy $S + U$ to zero:

$$\frac{\partial(S + U)}{\partial a} = 2\gamma - \frac{\sigma_f^2}{E}\pi a = 0$$

Since fast fracture is imminent when this condition is satisfied, we write the stress as σ_f . Solving,

$$\sigma_f = \sqrt{\frac{2E\gamma}{\pi a}}$$

Griffith's original work dealt with very brittle materials, specifically glass rods. When the material exhibits more ductility, consideration of the surface energy alone fails to provide an

accurate model for fracture. This deficiency was later remedied, at least in part, independently by Irwin⁴ and Orowan⁵. They suggested that in a ductile material a good deal – in fact the vast majority – of the released strain energy was absorbed not by creating new surfaces, but by energy dissipation due to plastic flow in the material near the crack tip. They suggested that catastrophic fracture occurs when the strain energy is released at a rate sufficient to satisfy the needs of all these energy “sinks,” and denoted this *critical strain energy release rate* by the parameter $\mathcal{G}_c$; the Griffith equation can then be rewritten in the form:

$$\sigma_f = \sqrt{\frac{E\mathcal{G}_c}{\pi a}} \quad (1)$$

This expression describes, in a very succinct way, the interrelation between three important aspects of the fracture process: the *material*, as evidenced in the critical strain energy release rate $\mathcal{G}_c$; the *stress level* σ_f ; and the *size*, a , of the flaw. In a design situation, one might choose a value of a based on the smallest crack that could be easily detected. Then for a given material with its associated value of $\mathcal{G}_c$, the safe level of stress σ_f could be determined. The structure would then be sized so as to keep the working stress comfortably below this critical value.

Example 1

The story of the DeHavilland Comet aircraft of the early 1950’s, in which at least two aircraft disintegrated in flight, provides a tragic but fascinating insight into the importance of fracture theory. It is an eerie story as well, having been all but predicted in a 1948 novel by Nevil Shute named *No Highway*. The book later became a movie starring James Stewart as a perserverant metallurgist convinced that his company’s new aircraft (the “Reindeer”) was fatally prone to metal fatigue. When just a few years later the Comet was determined to have almost exactly this problem, both the book and the movie became rather famous in the materials engineering community.

The postmortem study of the Comet’s problems was one of the most extensive in engineering history⁶. It required salvaging almost the entire aircraft from scattered wreckage on the ocean floor and also involved full-scale pressurization of an aircraft in a giant water tank. Although valuable lessons were learned, it is hard to overstate the damage done to the DeHavilland Company and to the British aircraft industry in general. It is sometimes argued that the long predominance of the United States in commercial aircraft is due at least in part to the Comet’s misfortune.

The Comet aircraft had a fuselage of clad aluminum, with $\mathcal{G}_c \approx 300$ in-psi. The hoop stress due to relative cabin pressurization was 20,000 psi, and at that stress the length of crack that will propagate catastrophically is

$$a = \frac{\mathcal{G}_c E}{\pi \sigma^2} = \frac{(300)(11 \times 10^6)}{\pi(20 \times 10^3)^2} = 2.62''$$

A crack would presumably be detected in routine inspection long before it could grow to this length. But in the case of the Comet, the cracks were propagating from rivet holes near the cabin windows. When the crack reached the window, the size of the window opening was effectively added to the crack length, leading to disaster.

Modern aircraft are built with this failure mode in mind, and have “tear strips” that are supposedly able to stop any rapidly growing crack. But this remedy is not always effective, as was demonstrated in 1988 when a B737 operated by Aloha Airlines had the roof of the first-class cabin tear away.. That aircraft had stress-corrosion damage at a number of rivets in the fuselage lap splices, and this permitted

⁴G.R. Irwin, “Fracture Dynamics,” *Fracturing of Metals*, American Society for Metals, Cleveland, 1948.

⁵E. Orowan, “Fracture and Strength of Solids,” Report of Progress in Physics, Vol. 12, 1949.

⁶T. Bishop, *Metal Progress*, Vol. 67, pp. 79–85, May 1955.

multiple small cracks to link up to form a large crack. A great deal of attention is currently being directed to protection against this sort of “multi-site damage.”

It is important to realize that the critical crack length is an absolute number, not depending on the size of the structure containing it. Each time the crack jumps ahead, say by a small increment δa , an additional quantity of strain energy is released from the newly-unloaded material near the crack. Again using our simplistic picture of a triangular-shaped region that is at zero stress while the rest of the structure continues to feel the overall applied stress, it is easy to see in Fig. 3 that much more more energy is released due to the jump at position 2 than at position 1. This is yet another reason why small things tend to be stronger: they simply aren’t large enough to contain a critical-length crack.

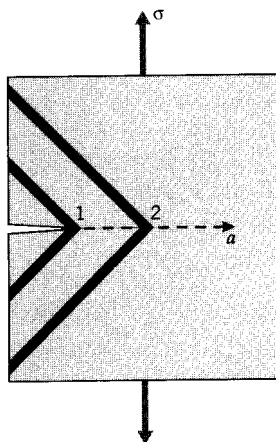


Figure 3: Energy released during an increment of crack growth, for two different crack lengths.

Example 2

Gordon⁷ tells of a ship’s cook who one day noticed a crack in the steel deck of his galley. His superiors assured him that it was nothing to worry about — the crack was certainly small compared with the vast bulk of the ship — but the cook began painting dates on the floor to mark the new length of the crack each time a bout of rough weather would cause it to grow longer. With each advance of the crack, additional decking material was unloaded, and the strain energy formerly contained in it released. But as the amount of energy released grows *quadratically* with the crack length, eventually enough was available to keep the crack growing even with no further increase in the gross load. When this happened, the ship broke into two pieces; this seems amazing but there are a more than a few such occurrences that are very well documented. As it happened, the part of the ship with the marks showing the crack’s growth was salvaged, and this has become one of the very best documented examples of slow crack growth followed by final catastrophic fracture.

Compliance calibration

A number of means are available by which the material property $\mathcal{G}_c$ can be measured. One of these is known as *compliance calibration*, which employs the concept of compliance as a ratio of

⁷J.E. Gordon, *Structures, or Why Things Don’t Fall Down*, Plenum, New York, 1978.

deformation to applied load: $C = \delta/P$. The total strain energy U can be written in terms of this compliance as:

$$U = \frac{1}{2}P\delta = \frac{1}{2}CP^2$$

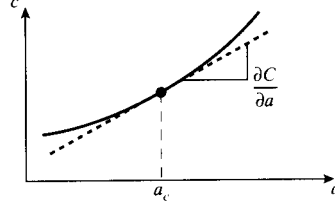


Figure 4: Compliance as a function of crack length.

The compliance of a suitable specimen, for instance a cantilevered beam, could be measured experimentally as a function of the length a of a crack that is grown into the specimen (see Fig. 4). The strain energy release rate can then be determined by differentiating the curve of compliance versus length:

$$\mathcal{G} = \frac{\partial U}{\partial a} = \frac{1}{2}P^2 \frac{\partial C}{\partial a} \quad (2)$$

The *critical* value of $\mathcal{G}$, $\mathcal{G}_c$, is then found by measuring the critical load P_c needed to fracture a specimen containing a crack of length a_c , and using the slope of the compliance curve at this same value of a :

$$\boxed{\mathcal{G}_c = \frac{1}{2}P_c^2 \frac{\partial C}{\partial a} \Big|_{a=a_c}} \quad (3)$$

Example 3

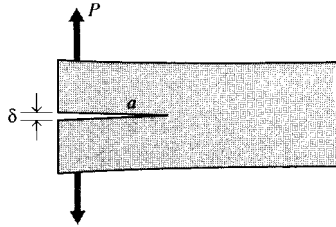


Figure 5: DCB fracture specimen.

For a double-cantilever beam (DCB) specimen such as that shown in Fig. 5, beam theory gives the deflection as

$$\frac{\delta}{2} = \frac{Pa^3}{3EI}$$

where $I = bh^3/12$. The elastic compliance is then

$$C = \frac{\delta}{P} = \frac{2a^3}{3EI}$$

If the crack is observed to jump forward when $P = P_c$, Eqn. 3 can be used to compute the critical strain energy release rate as

$$\mathcal{G}_c = \frac{1}{2}P_c^2 \cdot \frac{2a^2}{EI} = \frac{12P_c^2 a^2}{b^2 h^3 E}$$

The stress intensity approach

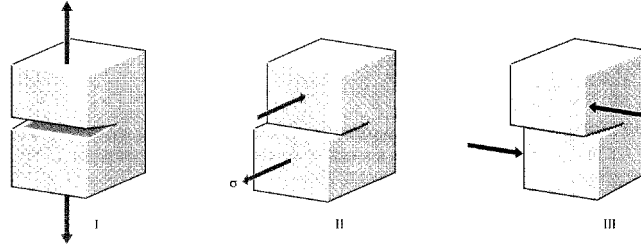


Figure 6: Fracture modes.

While the energy-balance approach provides a great deal of insight to the fracture process, an alternative method that examines the stress state near the tip of a sharp crack directly has proven more useful in engineering practice. The literature treats three types of cracks, termed mode I, II, and III as illustrated in Fig. 6. Mode I is a normal-opening mode and is the one we shall emphasize here, while modes II and III are shear sliding modes. As was outlined in Module 16, the semi-inverse method developed by Westergaard shows the opening-mode stresses to be:

$$\begin{aligned}\sigma_x &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) + \dots \\ \sigma_y &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 + \sin \frac{\theta}{2} \sin \frac{3\theta}{2} \right) + \dots \\ \tau_{xy} &= \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \cos \frac{3\theta}{2} \sin \frac{\theta}{2} \dots\end{aligned}\tag{4}$$

For distances close to the crack tip ($r \leq 0.1a$), the second and higher order terms indicated by dots may be neglected. At large distances from the crack tip, these relations cease to apply and the stresses approach their far-field values that would obtain were the crack not present.

The K_I in Eqns. 4 is a very important parameter known as the *stress intensity factor*. The I subscript is used to denote the crack opening mode, but similar relations apply in modes II and III. The equations show three factors that taken together depict the stress state near the crack tip: the denominator factor $(2\pi r)^{-1/2}$ shows the singular nature of the stress distribution; σ approaches infinity as the crack tip is approached, with a $r^{-1/2}$ dependency. The angular

dependence is separable as another factor; e.g. $f_x = \cos \theta/2 \cdot (1 - \sin \theta/2 \sin 3\theta/2) + \dots$. The factor K_I contains the dependence on applied stress σ_∞ , the crack length a , and the specimen geometry. The K_I factor gives the overall intensity of the stress distribution, hence its name.

For the specific case of a central crack of width $2a$ or an edge crack of length $2a$ in a large sheet, $K_I = \sigma_\infty \sqrt{\pi a}$, and $K_I = 1.12 \sigma_\infty \sqrt{\pi a}$ for an edge crack of length a in the edge of a large sheet. (The factor π could obviously be canceled with the π in the denominator of Eqn. 4, but is commonly retained for consistency with earlier work.) Expressions for K_I for some additional geometries are given in Table 1. The literature contains expressions for K for a large number of crack and loading geometries, and both numerical and experimental procedures exist for determining the stress intensity factor is specific actual geometries.

Table 1: Stress intensity factors for several common geometries.

Type of Crack	Stress Intensity Factor, K_I
Center crack, length $2a$, in an infinite plate	$\sigma_\infty \sqrt{\pi a}$
Edge crack, length a , in a semi-infinite plate	$1.12 \sigma_\infty \sqrt{\pi a}$
Central penny-shaped crack, radius a , in infinite body	$2 \sigma_\infty \sqrt{\frac{a}{\pi}}$
Center crack, length $2a$ in plate of width W	$\sigma_\infty \sqrt{W \tan \left(\frac{\pi a}{W} \right)}$
2 symmetrical edge cracks, each length a , in plate of total width W	$\sigma_\infty \sqrt{W \left[\tan \left(\frac{\pi a}{W} \right) + 0.1 \sin \left(\frac{2\pi a}{W} \right) \right]}$

These stress intensity factors are used in design and analysis by arguing that the material can withstand crack tip stresses up to a critical value of stress intensity, termed K_{Ic} , beyond which the crack propagates rapidly. This *critical stress intensity factor* is then a measure of material toughness. The failure stress σ_f is then related to the crack length a and the fracture toughness by

$$\sigma_f = \frac{K_{Ic}}{\alpha \sqrt{\pi a}} \quad (5)$$

where α is a geometrical parameter equal to 1 for edge cracks and generally on the order of unity for other situations. Expressions for α are tabulated for a wide variety of specimen and crack geometries, and specialty finite element methods are available to compute it for new situations.

The stress intensity and energy viewpoints are interrelated, as can be seen by comparing Eqns. 1 and 5 (with $\alpha = 1$):

$$\sigma_f = \sqrt{\frac{E\mathcal{G}_c}{\pi a}} = \frac{K_{Ic}}{\sqrt{\pi a}} \rightarrow K_{Ic}^2 = E\mathcal{G}_c$$

This relation applies in plane stress; it is slightly different in plane strain:

$$K_{Ic}^2 = E\mathcal{G}_c(1 - \nu^2)$$

For metals with $\nu = .3$, $(1 - \nu^2) = 0.91$. This is not a big change; however, the numerical values of $\mathcal{G}_c$ or K_{Ic} are very different in plane stress or plane strain situations, as will be described below.

Typical values of G_{Ic} and K_{Ic} for various materials are listed in Table 2, and it is seen that they vary over a very wide range from material to material. Some polymers can be very tough, especially when rated on a per-pound bases, but steel alloys are hard to beat in terms of absolute resistance to crack propagation.

Table 2: Fracture toughness of materials.

Material	$G_{Ic}(\text{kJm}^{-2})$	$K_{Ic}(\text{MNm}^2)$	$E(\text{GPa})$
Steel alloy	107	150	210
Aluminum alloy	20	37	69
Polyethylene	20 (J_{Ic})	—	0.15
High-impact polystyrene	15.8 (J_{Ic})	—	2.1
Steel — mild	12	50	210
Rubber	13	—	0.001
Glass-reinforced thermoset	7	7	7
Rubber-toughened epoxy	2	2.2	2.4
PMMA	0.5	1.1	2.5
Polystyrene	0.4	1.1	3
Wood	0.12	0.5	2.1
Glass	0.007	0.7	70

Example 4

Equation 5 provides a design relation among the applied stress σ , the material's toughness K_{Ic} , and the crack length a . Any one of these parameters can be calculated once the other two are known. To illustrate one application of the process, say we wish to determine the safe operating pressure in an aluminum pressure vessel 0.25 m in diameter and with a 5 mm wall thickness. First assuming failure by yield when the hoop stress reaches the yield stress (330 MPa) and using a safety factor of 0.75, we can compute the maximum pressure as

$$p = \frac{0.75\sigma t}{r} = \frac{0.75 \times 330 \times 10^6}{0.25/2} = 9.9 \text{ MPa} = 1400 \text{ psi}$$

To insure against failure by rapid crack growth, we now calculate the maximum crack length permissible at the operating stress, using a toughness value of $K_{Ic} = 41 \text{ MPa}\sqrt{\text{m}}$:

$$a = \frac{K_{Ic}^2}{\pi\sigma^2} = \frac{(41 \times 10^6)^2}{\pi(0.75 \times 330 \times 10^6)^2} = 0.01 \text{ m} = 0.4 \text{ in}$$

Here an edge crack with $\alpha = 1$ has been assumed. An inspection schedule must be implemented that is capable of detecting cracks before they reach this size.

Effect of specimen geometry

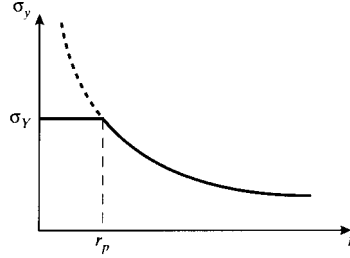


Figure 7: Stress limited by yield within zone r_p .

The toughness, or resistance to crack growth, of a material is governed by the energy absorbed as the crack moves forward. In an extremely brittle material such as window glass, this energy is primarily just that of rupturing the chemical bonds along the crack plane. But as already mentioned, in tougher materials bond rupture plays a relatively small role in resisting crack growth, with by far the largest part of the fracture energy being associated with plastic flow near the crack tip. A “plastic zone” is present near the crack tip within which the stresses as predicted by Eqn. 4 would be above the material’s yield stress σ_Y . Since the stress cannot rise above σ_Y , the stress in this zone is σ_Y rather than that given by Eqn. 4. To a first approximation, the distance r_p this zone extends along the x -axis can be found by using Eqn. 4 with $\theta = 0$ to find the distance at which the crack tip stress reduces to σ_Y :

$$\begin{aligned}\sigma_y = \sigma_Y &= \frac{K_I}{\sqrt{2\pi r_p}} \\ r_p &= \frac{K_I^2}{2\pi\sigma_Y^2}\end{aligned}\tag{6}$$

This relation is illustrated in Fig. 7. As the stress intensity is increased either by raising the imposed stress or by crack lengthening, the plastic zone size will increase as well. But the extent of plastic flow is ultimately limited by the material’s molecular or microstructural mobility, and the zone can become only so large. When the zone can grow no larger, the crack can no longer be constrained and unstable propagation ensues. The value of K_I at which this occurs can then be considered a materials property, named K_{Ic} .

In order for the measured value of K_{Ic} to be valid, the plastic zone size should not be so large as to interact with the specimen’s free boundaries or to destroy the basic nature of the singular stress distribution. The ASTM specification for fracture toughness testing⁸ specifies the specimen geometry to insure that the specimen is large compared to the crack length and the plastic zone size (see Fig. 8):

$$a, B, (W - a) \geq 2.5 \left(\frac{K_I}{\sigma_Y} \right)^2$$

⁸E 399-83, “Standard Test Method for Plane-Strain Fracture Toughness of Metallic Materials,” ASTM, 1983.

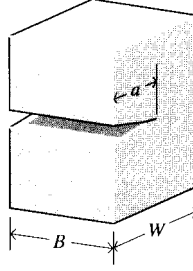


Figure 8: Dimensions of fracture toughness specimen.

A great deal of attention has been paid to the important case in which enough ductility exists to make it impossible to satisfy the above criteria. In these cases the stress intensity view must be abandoned and alternative techniques such as the J-integral or the crack tip opening displacement method used instead. The reader is referred to the references listed at the end of the module for discussion of these approaches.

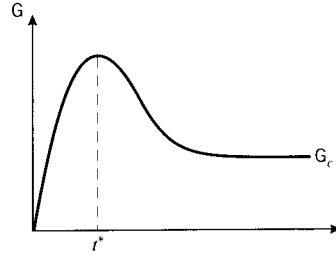


Figure 9: Effect of specimen thickness on toughness.

The fracture toughness as measured by K_c or $\mathcal{G}_c$ is essentially a measure of the extent of plastic deformation associated with crack extension. The quantity of plastic flow would be expected to scale linearly with the specimen thickness, since reducing the thickness by half would naturally cut the volume of plastically deformed material approximately in half as well. The toughness therefore rises linearly, at least initially, with the specimen thickness as seen in Fig. 9. Eventually, however, the toughness is observed to go through a maximum and fall thereafter to a lower value. This loss of toughness beyond a certain critical thickness t^* is extremely important in design against fracture, since using too thin a specimen in measuring toughness will yield an unrealistically optimistic value for $\mathcal{G}_C$. The specimen size requirements for valid fracture toughness testing are such that the most conservative value is measured.

The critical thickness is that which causes the specimen to be dominated by a state of *plane strain*, as opposed to *plane stress*. The stress in the through-thickness z direction must become zero at the sides of the specimen since no traction is applied there, and in a thin specimen the stress will not have room to rise to appreciable values within the material. The *strain* in the z direction is not zero, of course, and the specimen will experience a Poisson contraction given by $\epsilon_z = \nu(\sigma_x + \sigma_y)$. But when the specimen is thicker, material near the center will be unable to contract laterally due to the constraint of adjacent material. Now the z -direction strain is zero, so a tensile stress will arise as the material tries to contract but is prevented from doing so. The value of σ_z rises from zero at the outer surface and approaches a maximum value given

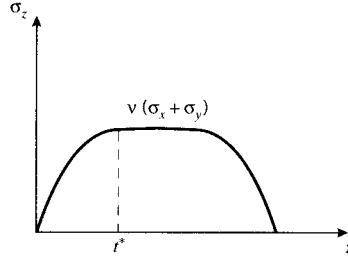


Figure 10: Transverse stress at crack tip.

by $\sigma_z \approx \nu(\sigma_x + \sigma_y)$ in a distance t^* as seen in Fig. 10. To guarantee that plane strain conditions dominate, the specimen thickness t must be such that $t \gg 2t^*$.

The triaxial stress state set up near the center of a thick specimen near the crack tip *reduces* the maximum shear stress available to drive plastic flow, since the maximum shear stress is equal to one half the difference of the largest and smallest principal stress, and the smallest is now greater than zero. Or equivalently, we can state that the mobility of the material is constrained by the inability to contract laterally. From either a stress or a strain viewpoint, the extent of available plasticity is reduced by making the specimen thick.

Example 5

The plastic zone sizes for the plane stress and plane strain cases can be visualized by using a suitable yield criterion along with the expressions for stress near the crack tip. The v. Mises yield criterion was given in terms of principal stresses in Module 20 as

$$2\sigma_Y^2 = (\sigma_1 - \sigma_2)^2 + (\sigma_1 - \sigma_3)^2 + (\sigma_2 - \sigma_3)^2$$

The principal stresses can be obtained from Eqns. 4 as

$$\sigma_1 = \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 + \sin \frac{\theta}{2} \right)$$

$$\sigma_2 = \frac{K_I}{\sqrt{2\pi r}} \cos \frac{\theta}{2} \left(1 - \sin \frac{\theta}{2} \right)$$

The third principal stress is

$$\sigma_3 = \begin{cases} 0, & \text{plane stress} \\ \nu(\sigma_1 + \sigma_2), & \text{plane strain} \end{cases}$$

These stresses can be substituted into the yield criterion, which is then solved for the radius r at which yield occurs. It is convenient to normalize this radius by the radius of the plastic zone along the x -axis, given by Eqn. 6. Maple commands to carry out these substitutions and plot the result are:

```
# Radius of plastic zone along x-axis
> rp:=K[I]^2/(2*Pi*sigma[Y]^2):

# v. Mises yield criterion in terms of principal stresses
> v_mises:=2*sigma[Y]^2= (sigma[1]-sigma[2])^2 + (sigma[1]-sigma[3])^2
+ (sigma[2]-sigma[3])^2:

# Principal stresses in crack-tip region
> sigma[1]:=(K[I]/sqrt(2*Pi*r))*cos(theta/2)*(1+sin(theta/2)):
```

```

> sigma[2] := (K[I]/sqrt(2*Pi*r))*cos(theta/2)*(1-sin(theta/2));

# Evaluate v. Mises for plane stress (v_strs) and plane strain (v_strn)
# Take nu = 0.3
> v_strs:=subs(sigma[3]=0,v_mises):
> v_strn:=subs(sigma[3]=.3*(sigma[1]+sigma[2]),v_mises):

# Solve for plastic zone radius, normalize by rp
# pl_strs for plane stress case, pl_strn for plane strain
> pl_strs:=solve(v_strs,r)/rp:
> pl_strn:=solve(v_strn,r)/rp:

# Plot normalized plastic zones for plane stress and plane strain
> plot({pl_strs,pl_strn},theta=0..2*Pi,coords=polar);

```

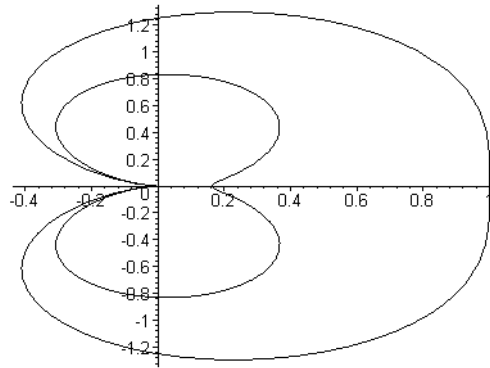


Figure 11: Normalized plastic zone shapes for plane strain (inner contour) and plane stress (outer contour).

Even in a thick specimen, the z -direction stress must approach zero at the side surfaces. Regions near the surface are therefore free of the triaxial stress constraint, and exhibit greater shear-driven plastic flow. After a cracked specimen has been tested to failure, a flat “thumbnail” pattern will often be visible as illustrated in Fig. 12. This is the region of slow crack growth, where the crack is able to maintain its preferred orientation transverse to the y -direction stress. The crack growth near the edges is retarded by the additional plastic flow there, so the crack line bows inward. When the stress is increased enough to cause the crack to grow catastrophically, it typically does so at speeds high enough that the transverse orientation is not always maintained. The region of rapid fracture is thus faceted and rough, leading some backyard mechanics to claim the material failed because it “crystallized.”

Along the edges of the specimen, “shear lips” can often be found on which the crack has developed by shear flow and with intensive plastic deformation. The lips will be near a 45° angle, the orientation of the maximum shear planes.

Grain size and temperature

Steel is such an important and widely used structural material that it is easy to forget that steel is a fairly recent technological innovation. Well into the nineteenth century, wood was the

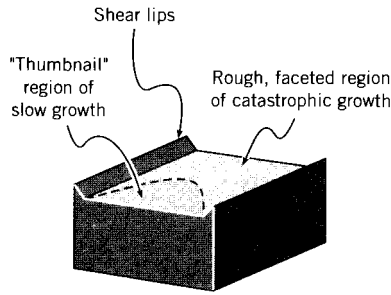


Figure 12: Fracture surface topography.

dominant material for many bridges, buildings, and ships. As the use of iron and steel became more widespread in the latter part of that century and the first part of the present one, a number of disasters took place that can be traced to the then-incomplete state of understanding of these materials, especially concerning their tendency to become brittle at low temperatures. Many of these failures have been described and analyzed in a fascinating book by Parker⁹.

One of these brittle failures is perhaps the most famous disaster of the last several centuries, the sinking of the transatlantic ocean liner *Titanic* on April 15, 1912, with a loss of some 1,500 people and only 705 survivors. Until very recently, the tragedy was thought to be caused by a long gash torn through the ship's hull by an iceberg. However, when the wreckage of the ship was finally discovered in 1985 using undersea robots, no evidence of such a gash was found. Further, the robots were later able to return samples of the ship's steel whose analysis has given rise to an alternative explanation.

It is now well known that lesser grades of steel, especially those having large concentrations of impurities such as interstitial carbon inclusions, are subject to embrittlement at low temperatures. William Garzke, a naval architect with the New York firm of Gibbs & Cox, and his colleagues have argued that the steel in the *Titanic* was indeed brittle in the 31°F waters of the Atlantic that night, and that the 22-knot collision with the iceberg generated not a gash but extensive cracking through which water could enter the hull. Had the steel remained tough at this temperature, these authors feel, the cracking may have been much less extensive. This would have slowed the flooding and allowed more time for rescue vessels to reach the scene, which could have increased greatly the number of survivors.

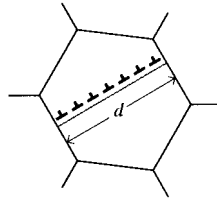


Figure 13: Dislocation pileup within a grain.

In the bcc transition metals such as iron and carbon steel, brittle failure can be initiated by dislocation glide within a crystalline grain. The slip takes place at the yield stress σ_Y , which

⁹E.R. Parker, *Brittle Behavior of Engineering Structures*, John Wiley & Sons, 1957.

varies with grain size according to the Hall-Petch law as described in Module 21:

$$\sigma_Y = \sigma_0 + k_Y d^{-1/2}$$

Dislocations are not able to propagate beyond the boundaries of the grain, since adjoining grains will not in general have their slip planes suitably oriented. The dislocations then “pile up” against the grain boundaries as illustrated in Fig. 13. The dislocation pileup acts similarly to an internal crack with a length that scales with the grain size d , intensifying the stress in the surrounding grains. Replacing a by d in the modified Griffith equation (Eqn. 1), the applied stress needed to cause fracture in adjacent grains is related to the grain size as

$$\sigma_f = k_f d^{-1/2}, \quad k_f \propto \sqrt{\frac{E\mathcal{G}_c}{\pi}}$$

The above two relations for yielding and fracture are plotted in Fig. 14 against inverse root grain size (so grain size increases to the left), with the slopes being k_Y and k_f respectively. When $k_f > k_Y$, fracture will not occur until $\sigma = \sigma_Y$ for values of d to the left of point A , since yielding and slip is a prerequisite for cleavage. In this region the yielding and fracture stresses are the same, and the failure appears brittle since large-scale yielding will not have a chance to occur. To the right of point A , yielding takes place prior to fracture and the material appears ductile. The point A therefore defines a critical grain size d^* at which a “nil-ductility” transition from ductile (grains smaller than d^*) to brittle failure will take place.

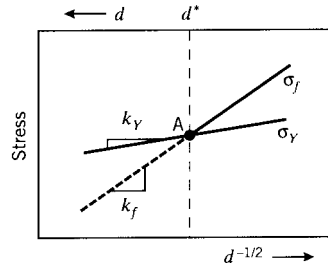


Figure 14: Effect of grain size on yield and fracture stress.

As the temperature is lowered, the yield stress σ_Y will increase as described in Module 20, and the fracture stress σ_f will decrease (since atomic mobility and thus $\mathcal{G}_C$ decrease). Therefore, point A shifts to the right as temperature is lowered. The critical grain size for nil ductility now occurs at a smaller value; i.e. the grains must be smaller to avoid embrittling the material. Equivalently, refining the grain size has the effect of lowering the ductile-brittle transition temperature. Hence grain-size refinement raises both the yield and fracture stress, lowers the ductile-brittle transition temperature, and promotes toughness as well. This is a singularly useful strengthening mechanism, since other techniques such as strain hardening and solid-solution hardening tend to achieve strengthening at the expense of toughness.

Factors other than temperature can also embrittle steel. Inclusions such as carbon and phosphorus act to immobilize slip systems that might otherwise relieve the stresses associated with dislocation pileups, and these inclusions can raise the yield stress and thus the ductile-brittle transition temperature markedly. Similar effects can be induced by damage from high-energy radiation, so embrittlement of nuclear reactor components is of great concern. Embrittlement is also facilitated by the presence of notches, since they generate triaxial stresses that constrain

plastic flow. High strain rates promote brittleness because the flow stress needed to accommodate the strain rate is higher, and improper welding can lead to brittleness both by altering the steel's microstructure and by generating residual internal stresses.

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Problems

1. Using a development analogous to that employed in Module 21 for the theoretical yield stress, show that the theoretical ultimate tensile strength is $\sigma_{th} \approx E/10$ (much larger than that observed experimentally). Assume a harmonic atomic force function $\sigma = \sigma_{th} \sin(2\pi x/\lambda)$, where x is the displacement of an atom from its equilibrium position and $\lambda \approx a_0$ is the interatomic spacing. The maximum stress σ_{th} can then be found by using

$$E = \left(\frac{d\sigma}{d\epsilon} \right)_{x \rightarrow 0} \quad \text{and} \quad \epsilon = \frac{x}{a_0}$$

2. Using a safety factor of 2, find the safe operating pressure in a closed-end steel pressure vessel 1' in diameter and 0.2" wall thickness.
3. A pressure vessel is constructed with a diameter of $d = 18''$ and a length of $L = 6'$. The vessel is to be capable of withstanding an internal pressure of $p = 1000$ psi, and the wall thickness is such as to keep the nominal hoop stress under 2500 psi. However, the vessel bursts at an internal pressure of only 500 psi, and a micrographic investigation reveals the

fracture to have been initiated by an internal crack 0.1" in length. Calculate the fracture toughness (K_{Ic}) of the material.

4. A highly cross-linked epoxy resin has a coefficient of linear thermal expansion $\alpha = 5 \times 10^{-5} \text{ K}^{-1}$, $G_{IC} = 120 \text{ J/m}^2$, $E = 3.2 \text{ GPa}$, and $\nu = 0.35$. A thick layer of resin is cured and is firmly bonded to an aluminum part ($\alpha = 2.5 \times 10^{-5} \text{ K}^{-1}$) at 180°C . Calculate the minimum defect size needed to initiate cracking in the resin on cooling to 20°C . Take α in Eqn. 5 to be $2/\pi$ for penny-shaped cracks of radius a in a wide sheet.
5. (a) A thick plate of aluminum alloy, 175 mm wide, contains a centrally-located crack 75 mm in length. The plate experiences brittle fracture at an applied stress (uniaxial, transverse to the crack) of 110 MPa. Determine the fracture toughness of the material.
(b) What would the fracture stress be if the plate were wide enough to permit an assumption of infinite width?
6. In order to obtain valid plane-strain fracture toughnesses, the plastic zone size must be small with respect to the specimen thickness B , the crack length a , and the "ligament" width $W - a$. The established criterion is

$$(W - a), B, a \geq \left(\frac{K_{Ic}}{\sigma_Y} \right)^2$$

Rank the materials in the database in terms of the parameter given on the right-hand side of this expression.

7. When a 150 kN load is applied to a tensile specimen containing a 35 mm crack, the overall displacement between the specimen ends is 0.5 mm. When the crack has grown to 37 mm, the displacement for this same load is 0.505 mm. The specimen is 40 mm thick. The fracture load of an identical specimen, but with a crack length of 36 mm, is 175 kN. Find the fracture toughness K_{Ic} of the material.

Fatigue

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Introduction

The concept of “fatigue” arose several times in the Module on Fracture (Module 23), as in the growth of cracks in the Comet aircraft that led to disaster when they became large enough to propagate catastrophically as predicted by the Griffith criterion. Fatigue, as understood by materials technologists, is a process in which damage accumulates due to the repetitive application of loads that may be well below the yield point. The process is dangerous because a single application of the load would not produce any ill effects, and a conventional stress analysis might lead to a assumption of safety that does not exist.

In one popular view of fatigue in metals, the fatigue process is thought to begin at an internal or surface flaw where the stresses are concentrated, and consists initially of shear flow along slip planes. Over a number of cycles, this slip generates intrusions and extrusions that begin to resemble a crack. A true crack running inward from an intrusion region may propagate initially along one of the original slip planes, but eventually turns to propagate transversely to the principal normal stress as seen in Fig. 1.

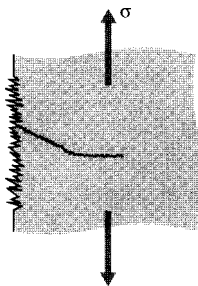


Figure 1: Intrusion-extrusion model of fatigue crack initiation.

When the failure surface of a fatigued specimen is examined, a region of slow crack growth is usually evident in the form of a “clamshell” concentric around the location of the initial flaw. (See Fig. 2.) The clamshell region often contains concentric “beach marks” at which the crack was arrested for some number of cycles before resuming its growth. Eventually, the crack may become large enough to satisfy the energy or stress intensity criteria for rapid propagation, following the previous expressions for fracture mechanics. This final phase produces the rough surface typical of fast fracture. In postmortem examination of failed parts, it is often possible to

correlate the beach marks with specific instances of overstress, and to estimate the applied stress at failure from the size of the crack just before rapid propagation and the fracture toughness of the material.



Figure 2: Typical fatigue-failure surfaces. From B. Chalmers, *Physical Metallurgy*, Wiley, p. 212, 1959.

The modern study of fatigue is generally dated from the work of A. Wöhler, a technologist in the German railroad system in the mid-nineteenth century. Wohler was concerned by the failure of axles after various times in service, at loads considerably less than expected. A railcar axle is essentially a round beam in four-point bending, which produces a compressive stress along the top surface and a tensile stress along the bottom (see Fig. 3). After the axle has rotated a half turn, the bottom becomes the top and vice versa, so the stresses on a particular region of material at the surface varies sinusoidally from tension to compression and back again. This is now known as *fully reversed* fatigue loading.

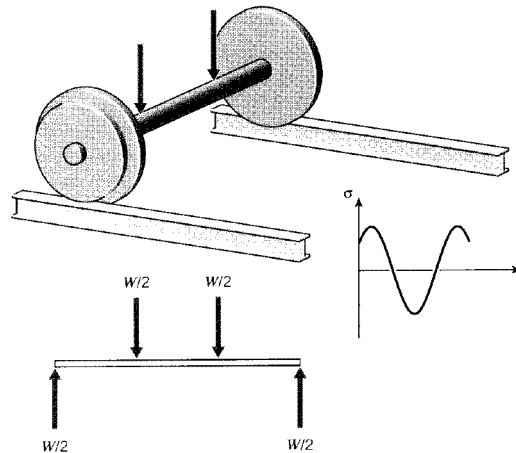


Figure 3: Fatigue in a railcar axle.

S-N curves

Well before a microstructural understanding of fatigue processes was developed, engineers had developed empirical means of quantifying the fatigue process and designing against it. Perhaps the most important concept is the S - N *diagram*, such as those shown in Fig. 4¹, in which a constant cyclic stress amplitude S is applied to a specimen and the number of loading cycles N until the specimen fails is determined. Millions of cycles might be required to cause failure at lower loading levels, so the abscissa is usually plotted logarithmically.

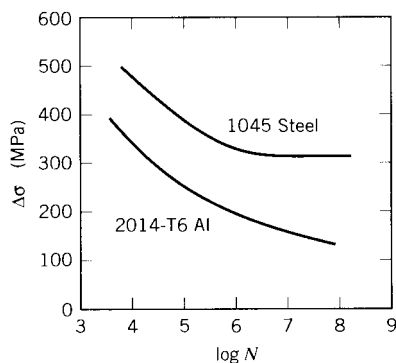


Figure 4: $S - N$ curves for aluminum and low-carbon steel.

In some materials, notably ferrous alloys, the $S - N$ curve flattens out eventually, so that below a certain *endurance limit* σ_e failure does not occur no matter how long the loads are cycled. Obviously, the designer will size the structure to keep the stresses below σ_e by a suitable safety factor if cyclic loads are to be withstood. For some other materials such as aluminum, no endurance limit exists and the designer must arrange for the planned lifetime of the structure to be less than the failure point on the $S - N$ curve.

Statistical variability is troublesome in fatigue testing; it is necessary to measure the lifetimes of perhaps twenty specimens at each of ten or so load levels to define the $S - N$ curve with statistical confidence². It is generally impossible to cycle the specimen at more than approximately 10Hz (inertia in components of the testing machine and heating of the specimen often become problematic at higher speeds) and at that speed it takes 11.6 days to reach 10^7 cycles of loading. Obtaining a full $S - N$ curve is obviously a tedious and expensive procedure.

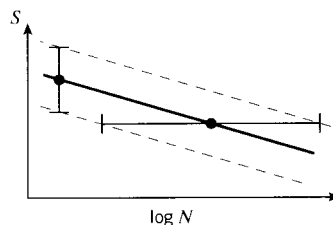


Figure 5: Variability in fatigue lifetimes and fracture strengths.

¹H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, John Wiley & Sons, 1965.

²*A Guide for Fatigue Testing and the Statistical Analysis of Fatigue Data*, ASTM STP-91-A, 1963.

At first glance, the scatter in measured lifetimes seems enormous, especially given the logarithmic scale of the abscissa. If the coefficient of variability in conventional tensile testing is usually only a few percent, why do the fatigue lifetimes vary over orders of magnitude? It must be remembered that in tensile testing, we are measuring the variability in stress at a given number of *cycles* (one), while in fatigue we are measuring the variability in cycles at a given *stress*. Stated differently, in tensile testing we are generating vertical scatter bars, but in fatigue they are horizontal (see Fig. 5). Note that we must expect more variability in the lifetimes as the $S - N$ curve becomes flatter, so that materials that are less prone to fatigue damage require more specimens to provide a given confidence limit on lifetime.

Effect of mean load

Of course, not all actual loading applications involve fully reversed stress cycling. A more general sort of fatigue testing adds a *mean stress* σ_m on which a sinusoidal cycle is superimposed, as shown in Fig. 6. Such a cycle can be phrased in several ways, a common one being to state the alternating stress σ_{alt} and the *stress ratio* $R = \sigma_{min}/\sigma_{max}$. For fully reversed loading, $R = -1$. A stress cycle of $R = 0.1$ is often used in aircraft component testing, and corresponds to a tension-tension cycle in which $\sigma_{min} = 0.1\sigma_{max}$.

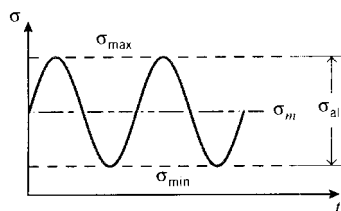


Figure 6: Simultaneous mean and cyclic loading.

A very substantial amount of testing is required to obtain an $S - N$ curve for the simple case of fully reversed loading, and it will usually be impractical to determine whole families of curves for every combination of mean and alternating stress. There are a number of strategems for finessing this difficulty, one common one being the *Goodman diagram*, shown in Fig. 7. Here a graph is constructed with mean stress as the abscissa and alternating stress as the ordinate, and a straight “lifeline” is drawn from σ_e on the σ_{alt} axis to the ultimate tensile stress σ_f on the σ_m axis. Then for any given mean stress, the endurance limit — the value of alternating stress at which fatigue fracture never occurs — can be read directly as the ordinate of the lifeline at that value of σ_m . Alternatively, if the design application dictates a given ratio of σ_e to σ_{alt} , a line is drawn from the origin with a slope equal to that ratio. Its intersection with the lifeline then gives the effective endurance limit for that combination of σ_f and σ_m .

Miner’s law for cumulative damage

When the cyclic load level varies during the fatigue process, a *cumulative damage* model is often hypothesized. To illustrate, take the lifetime to be N_1 cycles at a stress level σ_1 and N_2 at σ_2 . If damage is assumed to accumulate at a constant rate during fatigue and a number of cycles n_1 is applied at stress σ_1 , where $n_1 < N_1$ as shown in Fig. 8, then the fraction of lifetime consumed

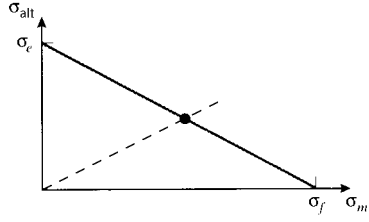


Figure 7: The Goodman diagram.

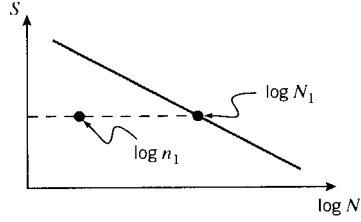


Figure 8: The concept of fractional lifetime.

will be n_1/N_1 . To determine how many additional cycles the specimen will survive at stress σ_2 , an additional fraction of life will be available such that the sum of the two fractions equals one:

$$\frac{n_1}{N_1} + \frac{n_2}{N_2} = 1$$

Note that absolute cycles and not log cycles are used here. Solving for the remaining cycles permissible at σ_2 :

$$n_2 = N_2 \left(1 - \frac{n_1}{N_1} \right)$$

The generalization of this approach is called *Miner's Law*, and can be written

$$\boxed{\sum \frac{n_j}{N_j} = 1} \quad (1)$$

where n_j is the number of cycles applied at a load corresponding to a lifetime of N_j .

Example 1

Consider a hypothetical material in which the S - N curve is linear from a value equal to the fracture stress σ_f at one cycle ($\log N = 0$), falling to a value of $\sigma_f/2$ at $\log N = 7$ as shown in Fig. 9. This behavior can be described by the relation

$$\log N = 14 \left(1 - \frac{S}{\sigma_f} \right)$$

The material has been subjected to $n_1 = 10^5$ load cycles at a level $S = 0.6\sigma_f$, and we wish to estimate how many cycles n_2 the material can now withstand if we raise the load to $S = 0.7\sigma_f$. From the S - N relationship, we know the lifetime at $S = 0.6\sigma_f = \text{constant}$ would be $N_1 = 3.98 \times 10^5$ and the lifetime at $S = 0.7\sigma_f = \text{constant}$ would be $N_2 = 1.58 \times 10^4$. Now applying Eqn. 1:

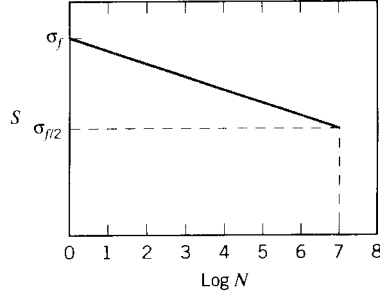


Figure 9: Linear S - N curve.

$$\frac{n_1}{N_1} + \frac{n_2}{N_2} = \frac{1 \times 10^5}{3.98 \times 10^5} + \frac{n_2}{1.58 \times 10^4} = 1$$

$$n_2 = 1.18 \times 10^4$$

Miner's "law" should be viewed like many other material "laws," a useful approximation, quite easy to apply, that *might* be accurate enough to use in design. But damage accumulation in fatigue is usually a complicated mixture of several different mechanisms, and the assumption of linear damage accumulation inherent in Miner's law should be viewed skeptically. If portions of the material's microstructure become unable to bear load as fatigue progresses, the stress must be carried by the surviving microstructural elements. The rate of damage accumulation in these elements then increases, so that the material suffers damage much more rapidly in the last portions of its fatigue lifetime. If on the other hand cyclic loads induce strengthening mechanisms such as molecular orientation or crack blunting, the rate of damage accumulation could drop during some part of the material's lifetime. Miner's law ignores such effects, and often fails to capture the essential physics of the fatigue process.

Crack growth rates

Certainly in aircraft, but also in other structures as well, it is vital that engineers be able to predict the rate of crack growth during load cycling, so that the part in question be replaced or repaired before the crack reaches a critical length. A great deal of experimental evidence supports the view that the crack growth rate can be correlated with the cyclic variation in the stress intensity factor³:

$$\boxed{\frac{da}{dN} = A \Delta K^m} \quad (2)$$

where da/dN is the fatigue crack growth rate per cycle, $\Delta K = K_{max} - K_{min}$ is the stress intensity factor range during the cycle, and A and m are parameters that depend the material, environment, frequency, temperature and stress ratio. This is sometimes known as the "Paris law," and leads to plots similar to that shown in Fig. 10.

³See Module 23.

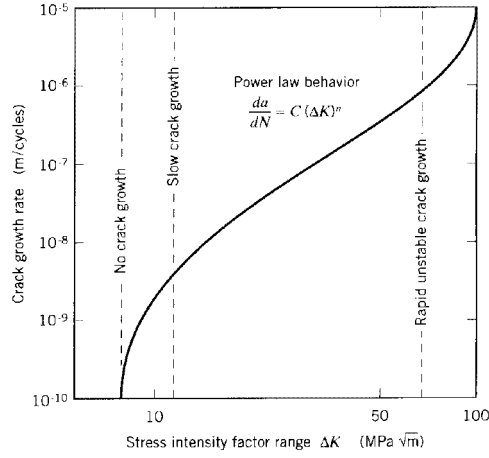


Figure 10: The Paris law for fatigue crack growth rates.

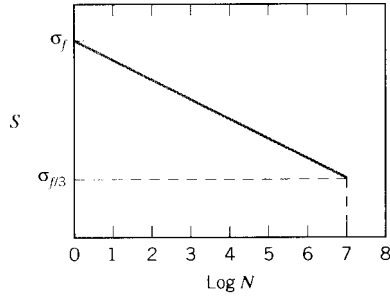
The exponent m is often near 4 for metallic systems, which might be rationalized as the damage accumulation being related to the volume V_p of the plastic zone: since the volume V_p of the zone scales with r_p^2 and $r_p \propto K_I^2$, then $da/dn \propto \Delta K^4$. Some specific values of the constants m and A for various alloys are given in Table 1.

Table 1: Numerical parameters in the Paris equation.

alloy	m	A
Steel	3	10^{-11}
Aluminum	3	10^{-12}
Nickel	3.3	4×10^{-12}
Titanium	5	10^{-11}

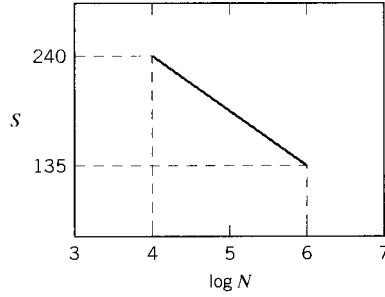
Problems

1. A steel has an ultimate tensile strength of 110 kpsi and a fatigue endurance limit of 50 kpsi. The load is such that the alternating stress is 0.4 of the mean stress. Using the Goodman method with a safety factor of 1.5, find the magnitude of alternating stress that gives safe operation.
2. A titanium alloy has an ultimate tensile strength of 120 kpsi and a fatigue endurance limit of 60 kpsi. The alternating stress is 20 kpsi. Find the allowable mean stress, using a safety factor of 2.
3. A material has an S - N curve that is linear from a value equal to the fracture stress σ_f at one cycle ($\log N = 0$), falling to a value of $\sigma_f/3$ at $\log N = 7$. The material has been subjected to $n_1 = 1000$ load cycles at a level $S = 0.7\sigma_f$. Estimate how many cycles n_2 the material can withstand if the stress amplitude is now raised to $S = 0.8\sigma_f$.



Prob. 3

4. A steel alloy has an S-N curve that falls linearly from 240 kpsi at 10^4 cycles to 135 kpsi at 10^6 cycles. A specimen is loaded at 160 kpsi alternating stress for 10^5 cycles, after which the alternating stress is raised to 180 kpsi. How many additional cycles at this higher stress would the specimen be expected to survive?



Prob. 4

5. Consider a body, large enough to be considered infinite in lateral dimension, containing a central through-thickness crack initially of length $2a_0$ and subjected to a cyclic stress of amplitude $\Delta\sigma$. Using the Paris Law (Eqn. 2), show that the number of cycles N_f needed for the crack to grow to a length $2a_f$ is given by the relation

$$\ln \left(\frac{a_f}{a_0} \right) = A (\Delta\sigma)^2 \pi N_f$$

when $m = 2$, and for other values of m

$$\left| a_f^{1-m/2} - a_0^{1-m/2} \right| = \frac{2-m}{2m} A (\Delta\sigma)^m \pi^{m/2} N_f$$

6. Use the expression obtained in Prob. 5 to compute the number of cycles a steel component can sustain before failure, where the initial crack halflength is 0.1 mm and the critical crack halflength to cause fracture is 2.5 mm. The stress amplitude per cycle is 950 MPa. Take the crack to be that of a central crack in an infinite plate.
7. Use the expression developed in Prob. 5 to investigate whether it is better to limit the size a_0 of initial flaws or to extend the size a_f of the flaw at which fast fracture occurs. Limiting a_0 might be done with improved manufacturing or better inspection methods,

and increasing a_f could be done by selecting a material with greater fracture toughness. For the “baseline” case, take $m = 3.5$, $a_0 = 2$ mm, $a_f =$ mm. Compute the percentage increase in N_f by letting (a) the initial faw size to be reduced to $a_0 = 1$ mm, and (b) increasing the final faw size to $N_f = 10$ mm.